SRM INSTITUTE OF SCIENCE & TECHNOLOGY,
RAMAPURAM – 600 089
FACULTY OF LIBERAL ARTS AND BUSINESS STUDIES
Faculty of Science & Humanities (A Place for Transformation)
Department of Computer Science W/S in Artificial Intelligence & Machine Learning

PRACTICAL RECORD

NAME :

REGISTER NUMBER :

PROGRAMME : **M.Sc. (Cyber Security)**

SEMESTER/YEAR : **1 / I**

SECTION : **A**

COURSE CODE : **PCS25C13J**

COURSE NAME : **COMPUTER NETWORKS**

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RAMAPURAM – 600 089
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REGISTER NUMBER:

BONAFIDE CERTIFICATE

This is to certify that the bonafide work done by _____in the subject Computer Networks (PCS25C13J) at, SRM Institute of Science and Technology, Chennai Ramapuram in October 2026.

STAFF IN-CHARGE **HEAD OF THE DEPARTMENT**

Submitted for the University Practical Examination held at SRM Institute of Science and Technology, Ramapuram Campus, Chennai on _____.

INTERNAL EXAMINER 1 **INTERNAL EXAMINER 2**

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Ex. No: 1 **Name:**

Date: **Reg. No.:**

Familiarization with Network Commands

Aim

To study and execute basic network commands and understand their functions.

Procedure

1. Open Command Prompt.
2. Execute the following commands one by one:
    - ipconfig
    - ping google.com
    - tracert google.com
    - nslookup google.com
    - netstat -an
    - arp -a
    - hostname
    - getmac
3. Observe and record the output.
4. Note the purpose of each command.

Output

IP Configuration

C:\> ipconfig

IPv4 Address : 192.168.1.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

Ping

C:\> ping google.com

Reply from 142.250.xxx.xxx:

Packets: Sent = 4, Received = 4, Lost = 0

Hostname

C:\> hostname

Result

The basic network commands were executed successfully, and the system's network configuration, connectivity, routing information, and active connections were observed.

Ex. No: 2 **Name:**
Date: **Reg. No.:**

Familiarization with Cisco Packet Tracer

Aim

To familiarize students with the Cisco Packet Tracer interface and create a simple network topology for testing communication.

Procedure

Step 1: Open Packet Tracer Launch Cisco Packet Tracer.

Step 2: Add Devices From the device list:

- 1 Switch (2960)
- 2 PCs

Drag and drop them into the workspace.

Step 3: Connect Devices

Choose
Connections
↓
Copper Straight Through Cable
Connect
PC0 FastEthernet0
↓
Switch Fa0/1

PC1 FastEthernet0
↓
Switch Fa0/2

Step 4: Configure IP Address

PC0

Desktop → IP Configuration

IP Address : 192.168.1.10

Subnet Mask: 255.255.255.0

Gateway : Blank

PC1

Desktop → IP Configuration

IP Address : 192.168.1.20

Subnet Mask: 255.255.255.0

Gateway : Blank

Step 5: Verify Connectivity On PC0 ,

Desktop → Command Prompt

ping 192.168.1.20

Expected Output:

Reply from 192.168.1.20

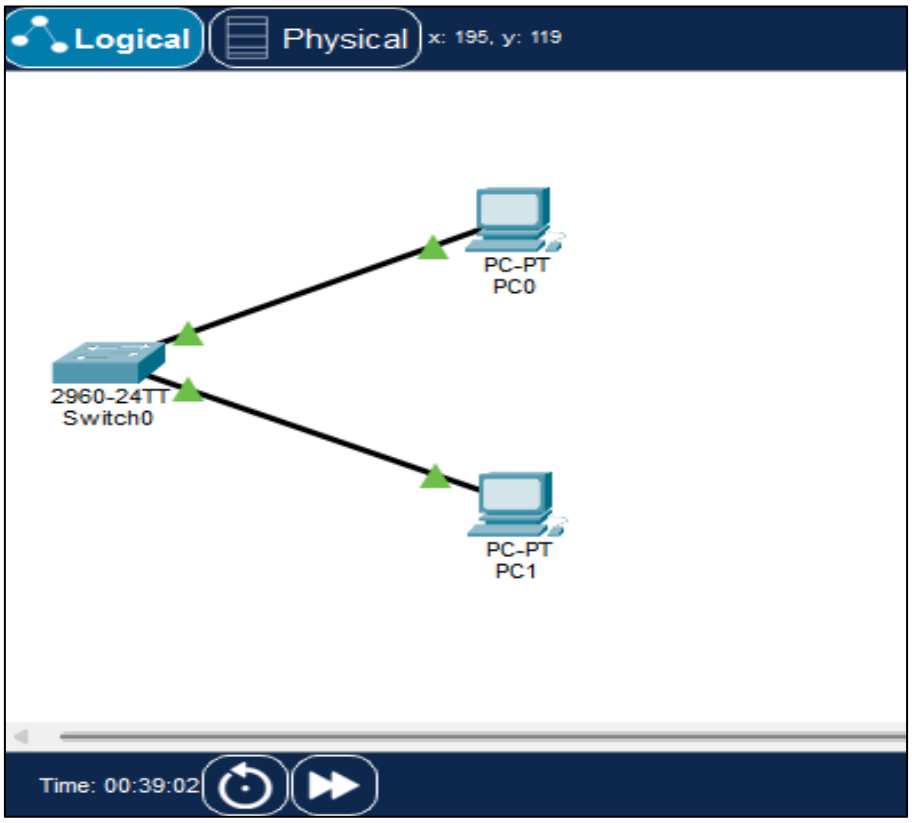
Packets: Sent = 4 Received = 4 Lost = 0

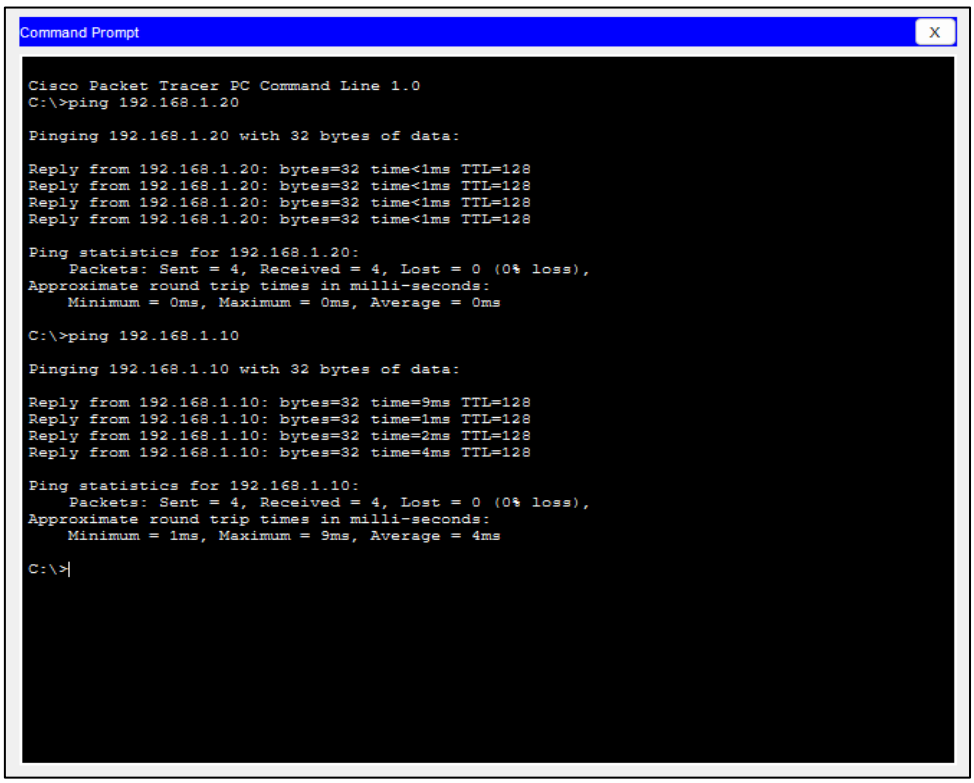
Step 6: Test Reverse ,

Ping From PC1 ping 192.168.1.10

Successful replies indicate proper communication.

Output





Result

The Cisco Packet Tracer environment was successfully explored. A simple LAN was created using two PCs and one switch. IP addresses were configured, and successful communication between the PCs was verified using the Ping command.

Ex. No: 3 **Name:**

Date: **Reg. No.:**

IP Addressing and Device Configuration Using Cisco Packet Tracer

Aim

To configure IPv4 addresses on end devices and a router using Cisco Packet Tracer and verify network connectivity.

Procedure

Step 1: Create the Network Add the following devices:

- 1 Router (Cisco 1941)
- 2 Switches (Cisco 2960)
- 2 PCs

Step 2: Connect the Devices

Use Copper Straight-Through cables.

- PC0 → Switch0
- Switch0 → Router0 (G0/0)
- Router0 (G0/1) → Switch1
- Switch1 → PC1

Step 3: Configure IP Address on PC0

Desktop → IP Configuration

IP Address : 192.168.10.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.10.1

Step 4: Configure IP Address on PC1

IP Address : 192.168.20.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.20.1

Program

Router Configuration:

Click Router0 → CLI

Configure Interface G0/0

Router> enable

Router# configure terminal

Router(config)# interface gigabitEthernet0/0

Router(config-if)# ip address 192.168.10.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Configure Interface G0/1:

Router(config)# interface gigabitEthernet0/1

Router(config-if)# ip address 192.168.20.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Router(config)# end

Router# copy running-config startup-config

Verification Commands:

Check Interface Status

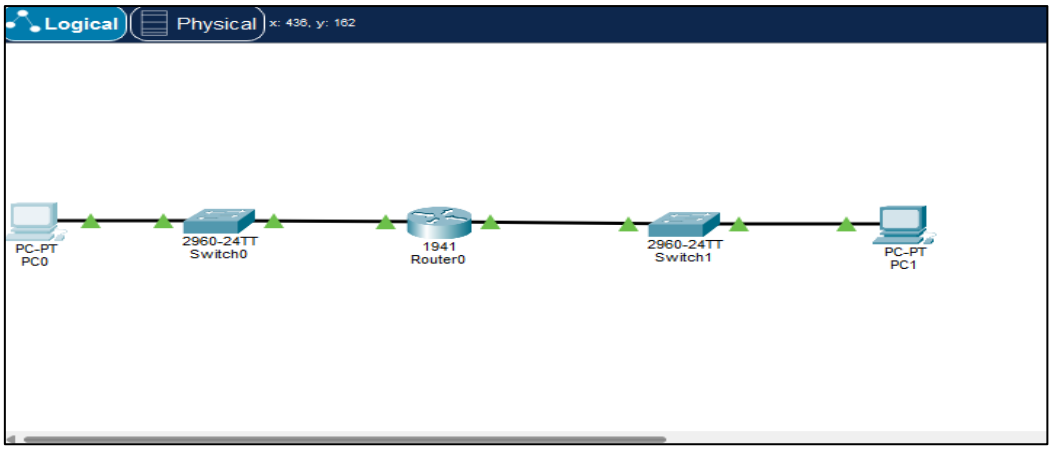
Router# show ip interface brief

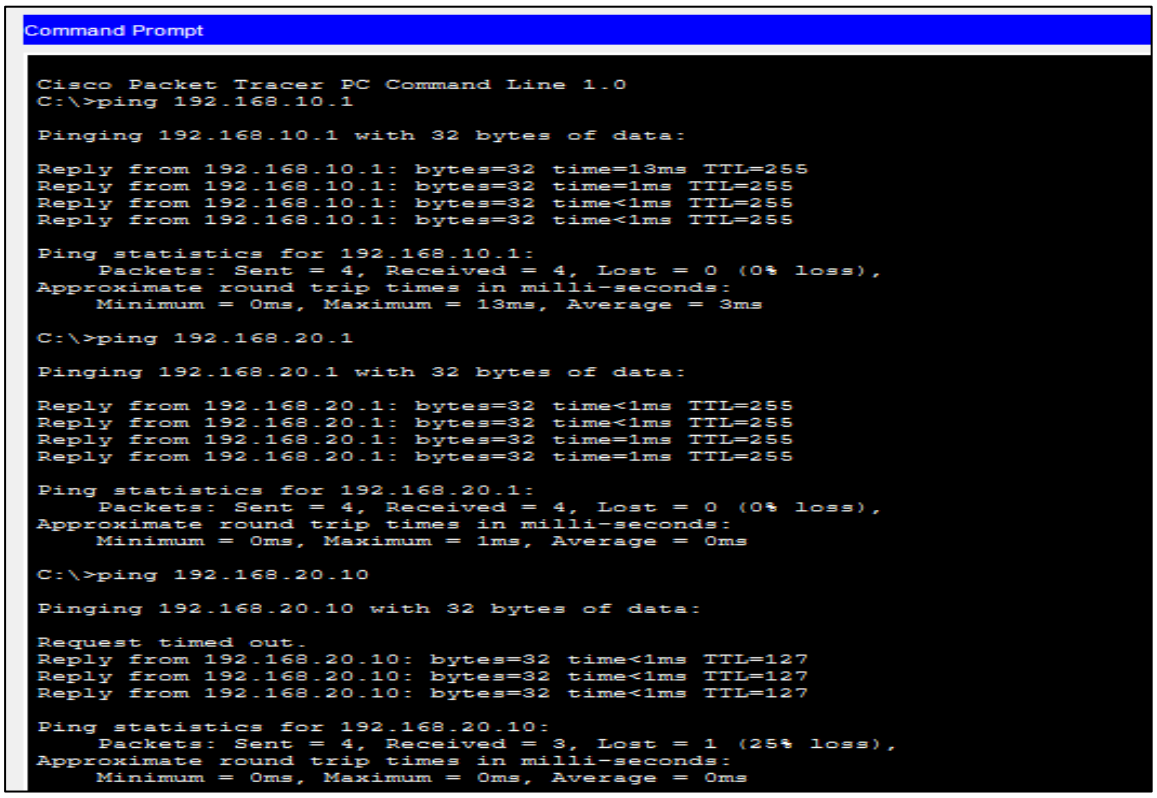
Expected Output

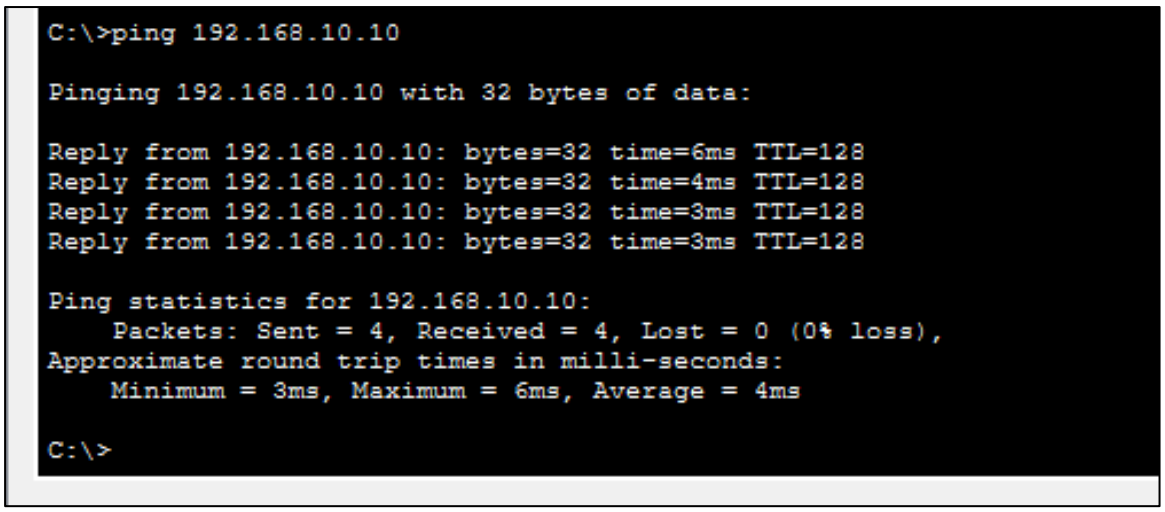
Interface	IP Address	Status
GigabitEthernet0/0	192.168.10.1	up
GigabitEthernet0/1	192.168.20.1	up

Output

Connectivity







Result

The IPv4 addresses were successfully configured on the PCs and router interfaces using Cisco Packet Tracer. The router interfaces were enabled, and end-to-end connectivity between different networks was verified successfully using the Ping command.

Ex. No: 4 **Name:**

Date: **Reg. No.:**

Establishing a Peer-to-Peer Network

Aim

To set up and configure a simple **peer-to-peer (P2P) LAN** in a computer lab so that all connected PCs can share resources directly without a central server

Procedure

1. Physical Setup

1. Power on all PCs.
2. Connect each PC to the hub/switch using Ethernet cables.
    - For direct PC-to-PC link, use a crossover cable
3. If internet is required, connect the router to the switch/hub

2. Assign IP Addresses

1. On each PC, open Control Panel → Network and Sharing Center → Change adapter settings.
2. Right-click the Ethernet adapter → Properties.
3. Select Internet Protocol Version 4 (TCP/IPv4) → Properties
4. Set Static IP for each PC (e.g., PC1: 192.168.1.2, PC2: 192.168.1.3),
    - Subnet mask: 255.255.255.0
    - Default gateway: Router IP (if used)

   DNS: Optional (leave blank or set to router IP)

Verify Connectivity

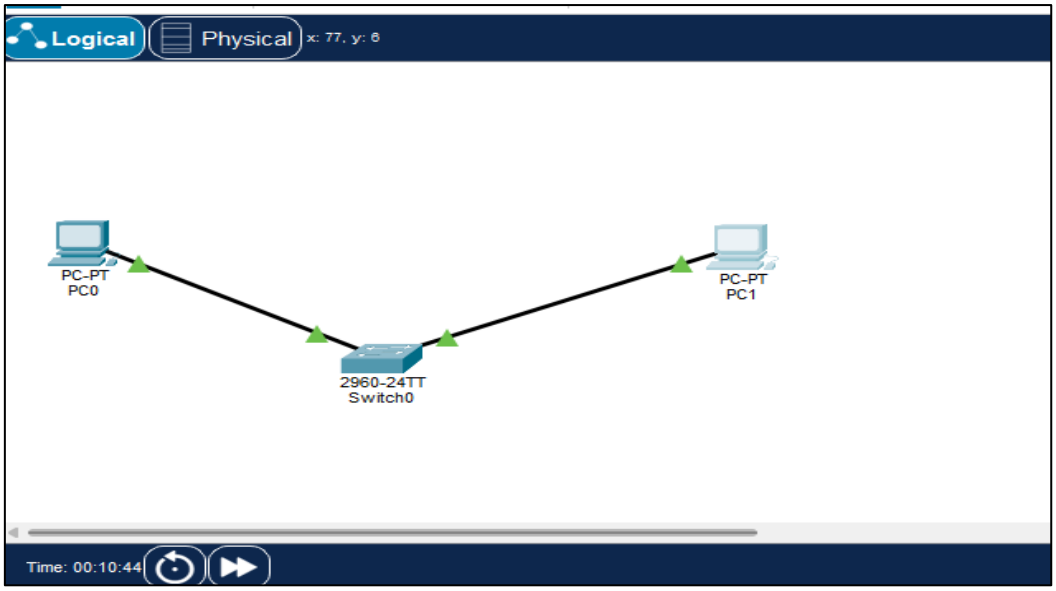
1. Open **Command Prompt** on each PC.
2. Run:

   ping 192.168.1.3 // from PC1 to PC2

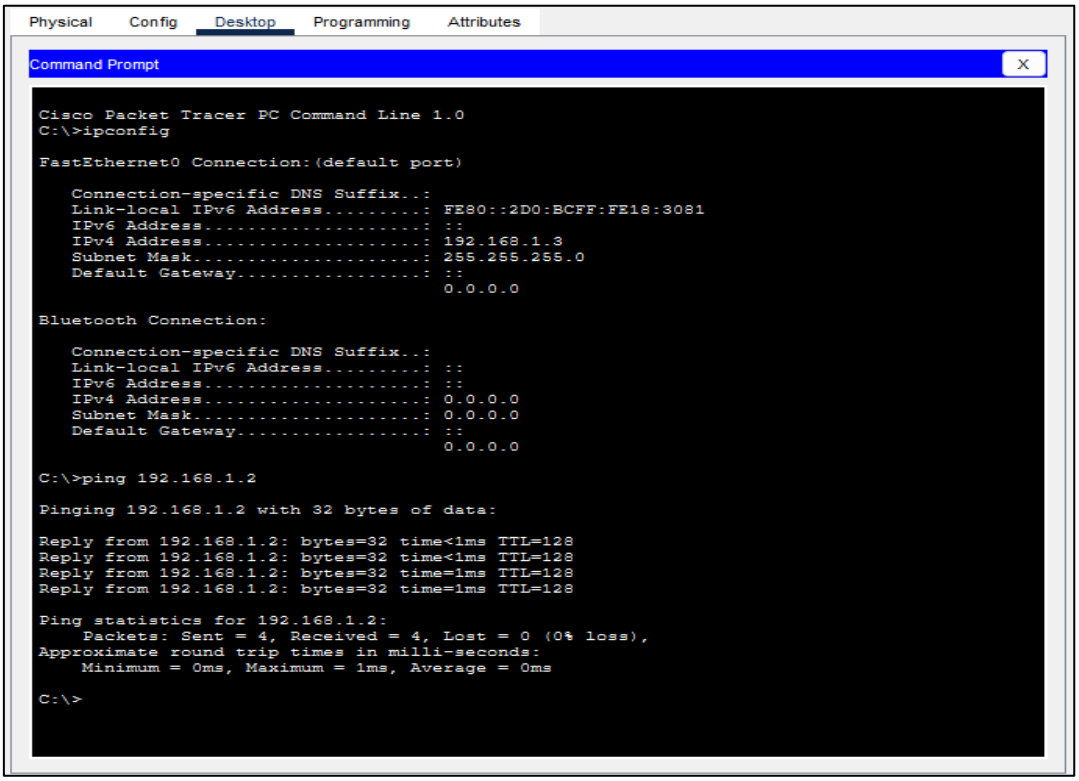
   ping 192.168.1.2 // from PC2 to PC1

Output

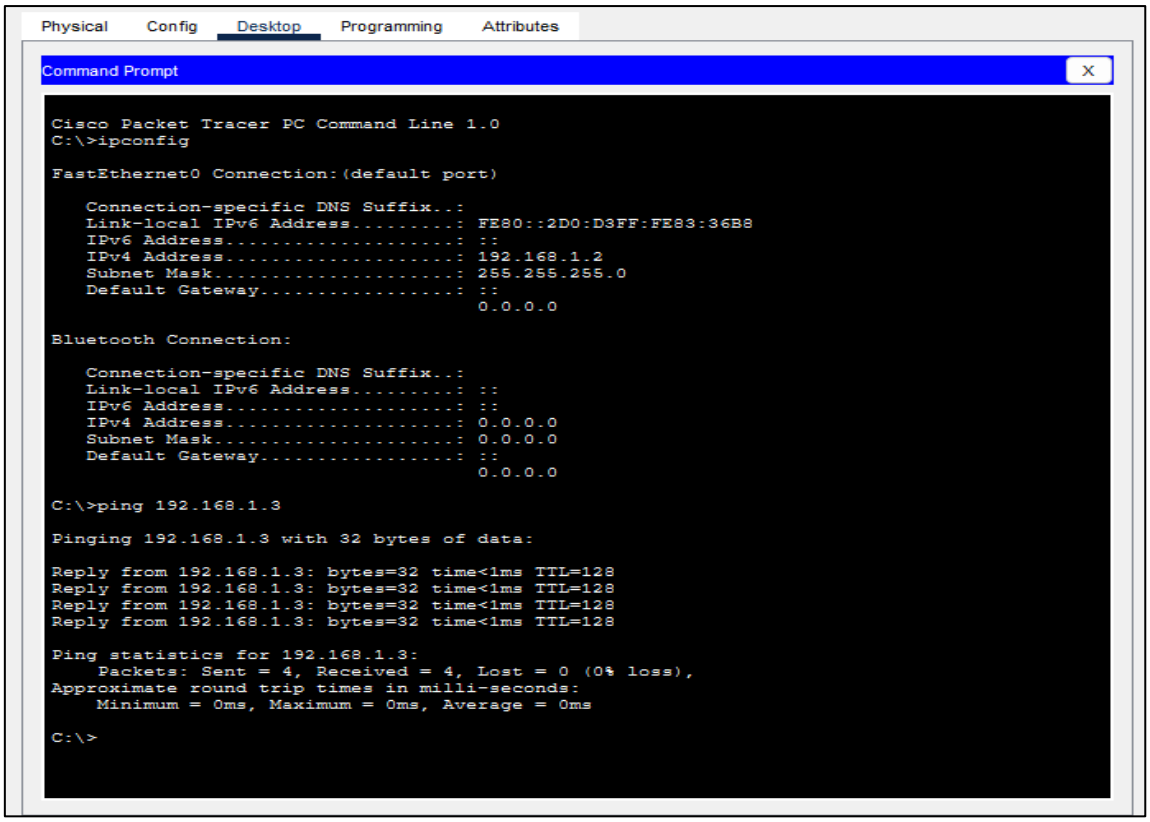
Connectivity



For PC1



For PC0



Result

To set up and configure a simple **peer-to-peer (P2P) LAN** in a computer lab so that all connected PCs can share resources directly without a central server has been executed successfully and verified.

Ex. No: 5 **Name:**

Date: **Reg. No.:**

Demonstrate the Working of LAN Topologies

Aim

To demonstrate the working of different LAN topologies using Cisco Packet Tracer and verify communication between network devices.

Algorithm

Step 1: Open Cisco Packet Tracer.

Step 2: Select the required network devices such as PCs, switches, and routers.

Step 3: Arrange the devices according to the selected LAN topology.

Step 4: Connect the devices using appropriate network cables.

Step 5: Assign unique IP addresses to all PCs.

Step 6: Configure the subnet mask for all devices.

Step 7: Verify the physical connections.

Step 8: Open the Command Prompt of a PC.

Step 9: Execute the ping command to test connectivity between PCs.

Step 10: Observe the packets and verify successful communication.

Step 11: Repeat the above steps for Star, Bus, Ring, Mesh, and Tree topologies.

Step 12: Record the observations and results.

Program / Configuration

Cisco Packet Tracer does not require a conventional programming language for this experiment. The following IP configuration and network commands are used.

PC1 Configuration

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.254

PC2 Configuration
IP Address : 192.168.1.2
Subnet Mask : 255.255.255.0
Default Gateway: 192.168.1.254

PC3 Configuration
IP Address : 192.168.1.3
Subnet Mask : 255.255.255.0
Default Gateway: 192.168.1.254

PC4 Configuration
IP Address : 192.168.1.4
Subnet Mask : 255.255.255.0
Default Gateway: 192.168.1.254

Connectivity Testing Command

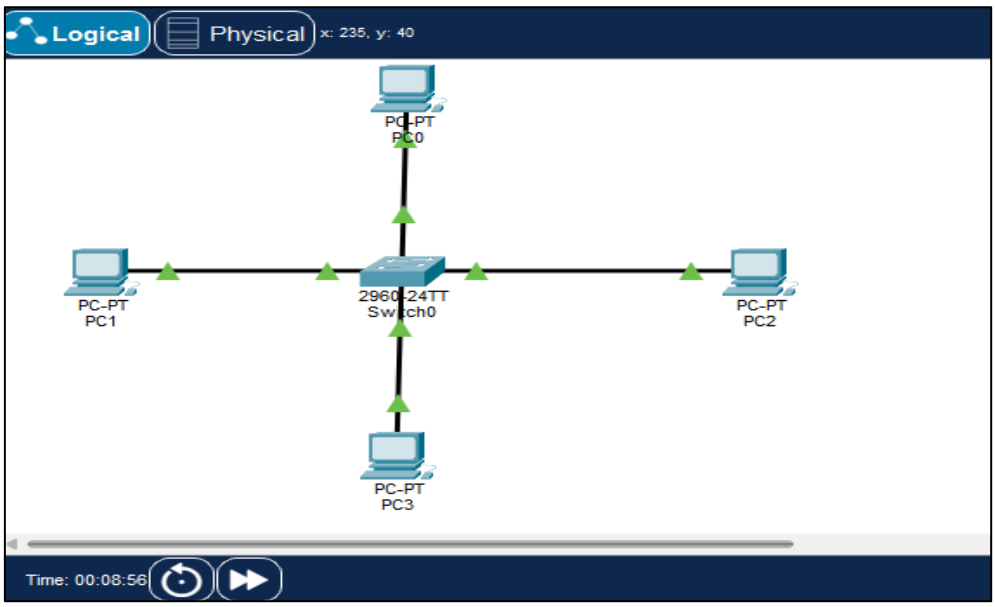
From PC1, open Command Prompt and execute:
ping 192.168.1.2
To test other PCs:
ping 192.168.1.3
ping 192.168.1.4

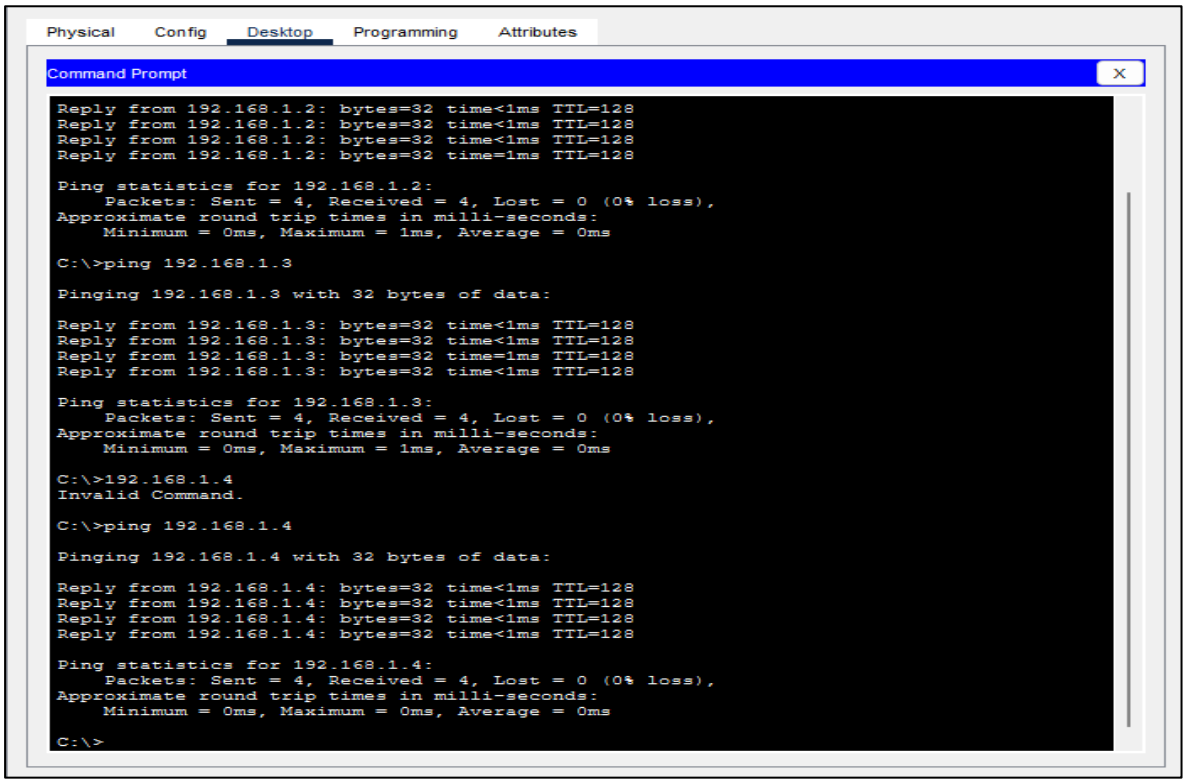
Input

- Number of PCs: 4
- Network topology: Star / Bus / Ring / Mesh / Tree
- IP addresses:
  - PC1 → 192.168.1.1
  - PC2 → 192.168.1.2
  - PC3 → 192.168.1.3
  - PC4 → 192.168.1.4
- Subnet Mask → 255.255.255.0

Output

Connectivity





Result

The different LAN topologies—Star, Bus, Ring, Mesh, and Tree—were successfully designed and demonstrated using Cisco Packet Tracer. The connectivity between the network devices was verified successfully using the ping command.

Ex. No: 6 **Name:**

Date: **Reg. No.:**

Analyze MAC Address Behavior

Aim

To analyze the behavior of MAC addresses in a LAN and observe how a switch learns and uses MAC addresses for forwarding Ethernet frames.

Algorithm

Step 1: Open Cisco Packet Tracer.

Step 2: Place a switch and three PCs in the workspace.

Step 3: Connect the PCs to the switch using copper straight-through cables.

Step 4: Assign IP addresses to the PCs.

Step 5: Open the Command Prompt of PC1.

Step 6: Use the ping command to communicate with PC2.

Step 7: Observe the MAC address table of the switch.

Step 8: Use the show mac address-table command on the switch.

Step 9: Record the MAC addresses learned by the switch.

Step 10: Send data between different PCs and observe the changes in the MAC address table.

Step 11: Analyze how the switch forwards frames based on destination MAC addresses.

Step 12: Record the observations and conclude the behavior of MAC addresses.

Program / Commands

PC1:

IP Address : 192.168.1.1

Subnet Mask : 255.255.255.0

PC2:

IP Address : 192.168.1.2

Subnet Mask : 255.255.255.0

PC3:

IP Address : 192.168.1.3

Subnet Mask : 255.255.255.0

Check MAC Address

On each PC:

ipconfig /all

or use:

arp -a

Test Connectivity

From PC1:

ping 192.168.1.2

From PC2:

ping 192.168.1.3

Display Switch MAC Address Table

On the switch CLI:

Switch> enable

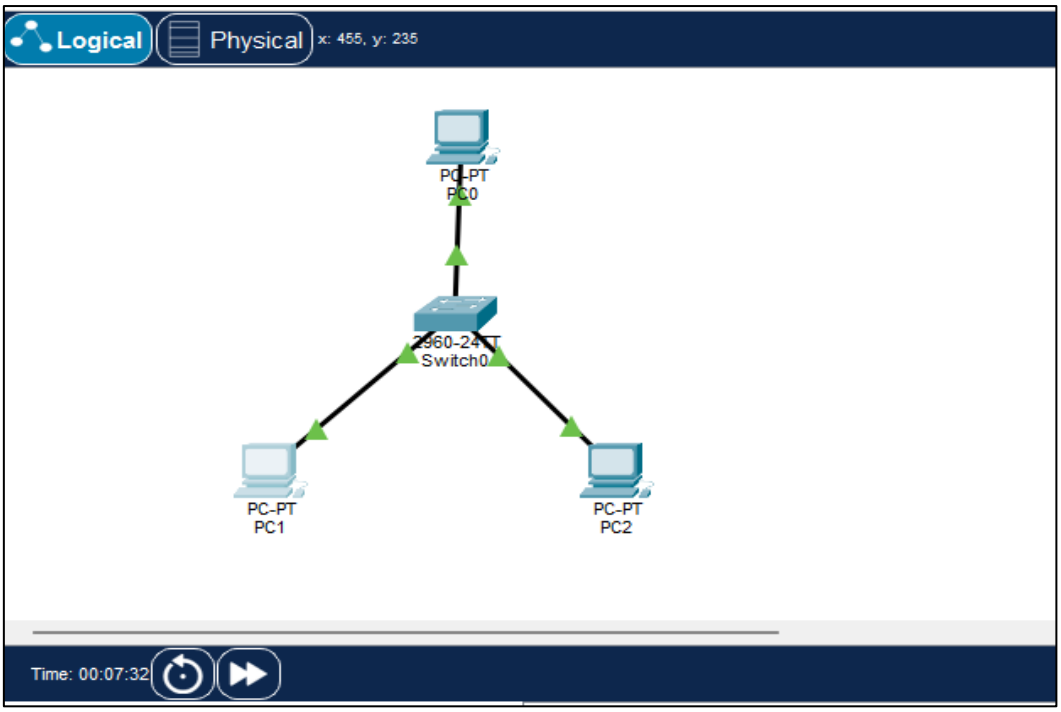
Switch# show mac address-table

Input

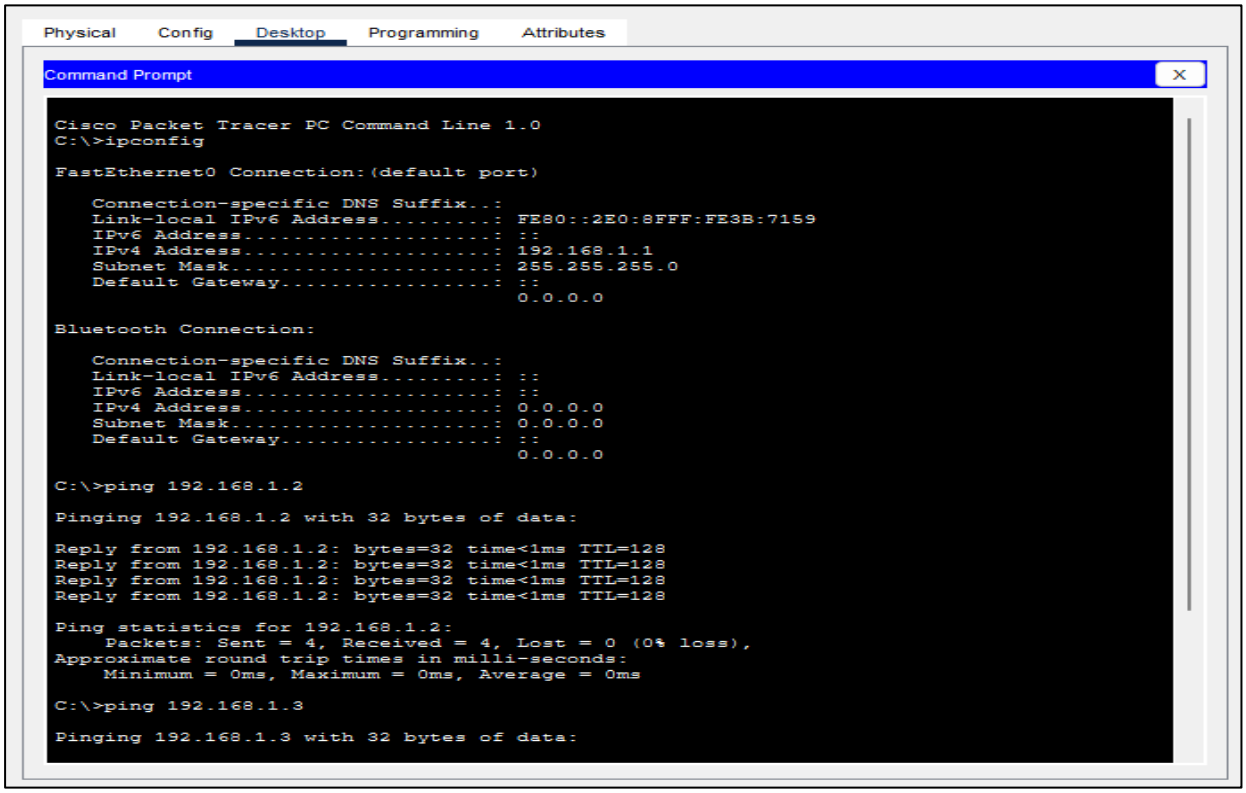
- Number of PCs: 3
- Network device: 1 Switch
- PC1 IP: 192.168.1.1
- PC2 IP: 192.168.1.2
- PC3 IP: 192.168.1.3
- Subnet Mask: 255.255.255.0
- Commands: ping, ipconfig /all, show mac address-table

Output

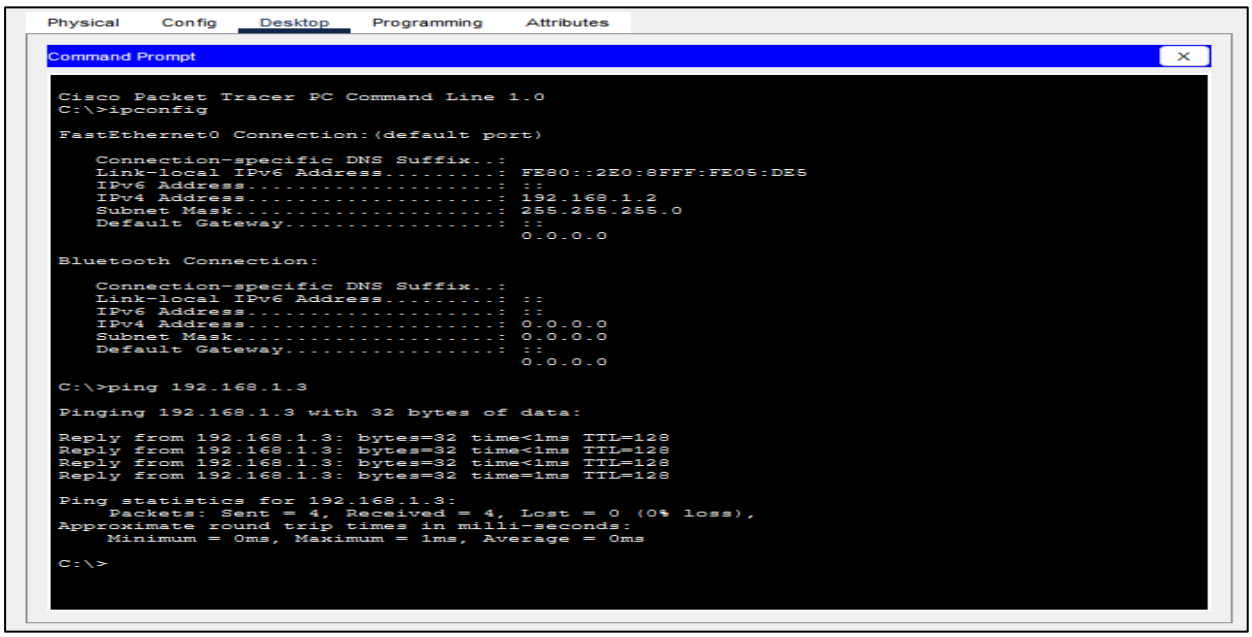
Connectivity



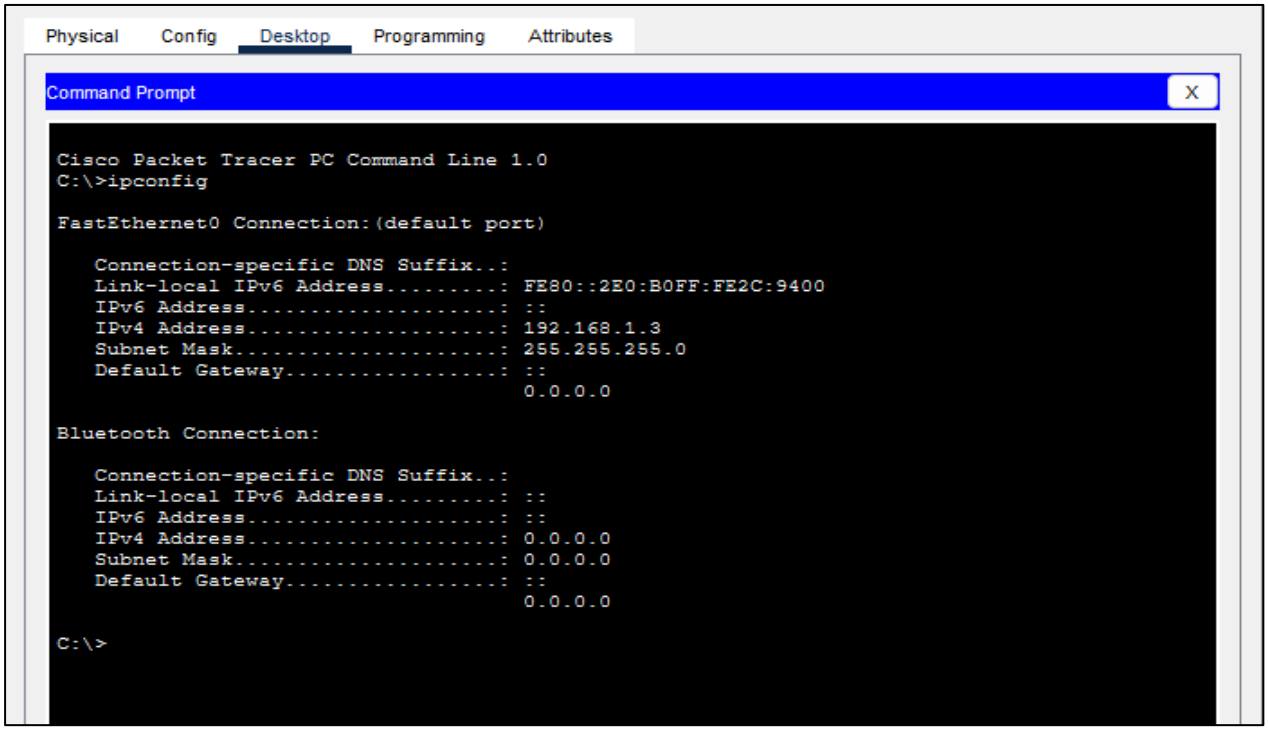
For PC0



For PC1



For PC2:



Result

The MAC address behavior in a LAN was successfully analyzed using Cisco Packet Tracer. The switch learned the MAC addresses of connected devices dynamically and used the MAC address table to forward Ethernet frames to the appropriate destination port.

Ex. No: 7 **Name:**

Date: **Reg. No.:**

Connecting Two LANs Using Static Routing

Aim

To connect two different LANs using two routers and configure static routing to enable communication between the networks.

Algorithm

STEP 1: Open Cisco Packet Tracer

STEP 2: Place two routers, two switches, and PCs for each LAN.

STEP 3: Connect PCs to their respective switches.

STEP 4: Connect each switch to a router.

STEP 5: Connect the two routers using a suitable cable.

STEP 6: Assign IP addresses to all PCs and router interfaces.

STEP 7: Configure the interfaces of both routers.

STEP 8: Add static routes in both routers using the ip route command.

STEP 9: Verify the router interfaces using show ip interface brief.

STEP 10: Test connectivity between PCs in different LANs using the ping command.

STEP 11: Observe the successful transmission of packets between the two LANs.

STEP 12: Record the output and result.

Network Configuration

LAN 1

Device	IP Address	Subnet Mask	Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.1.11	255.255.255.0	192.168.1.1

Router 1 LAN interface

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

LAN 2

Device	IP Address	Subnet Mask	Gateway
PC3	192.168.2.10	255.255.255.0	192.168.2.1
PC4	192.168.2.11	255.255.255.0	192.168.2.1

Router 2 LAN interface:

IP Address: 192.168.2.1

Subnet Mask: 255.255.255.0

Router-to-Router Connection

Router 1: 10.0.0.1 /30

Router 2: 10.0.0.2 /30

Subnet Mask: 255.255.255.252 4.

Program

Router 1

```
Router> enable

Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface serial 0/0/0
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

```
Router(config)# end
Router# show ip route
```

Router 2

```
Router> enable

Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface serial 0/0/0
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit

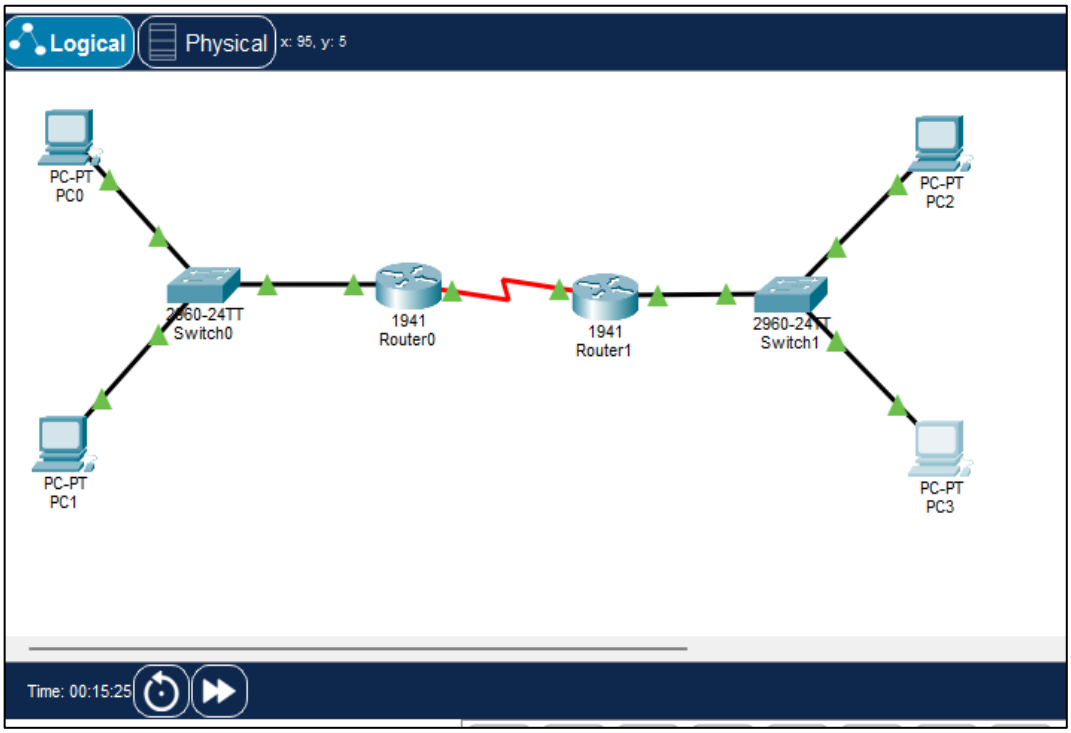
Router(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)# end
Router# show ip route
```

Input

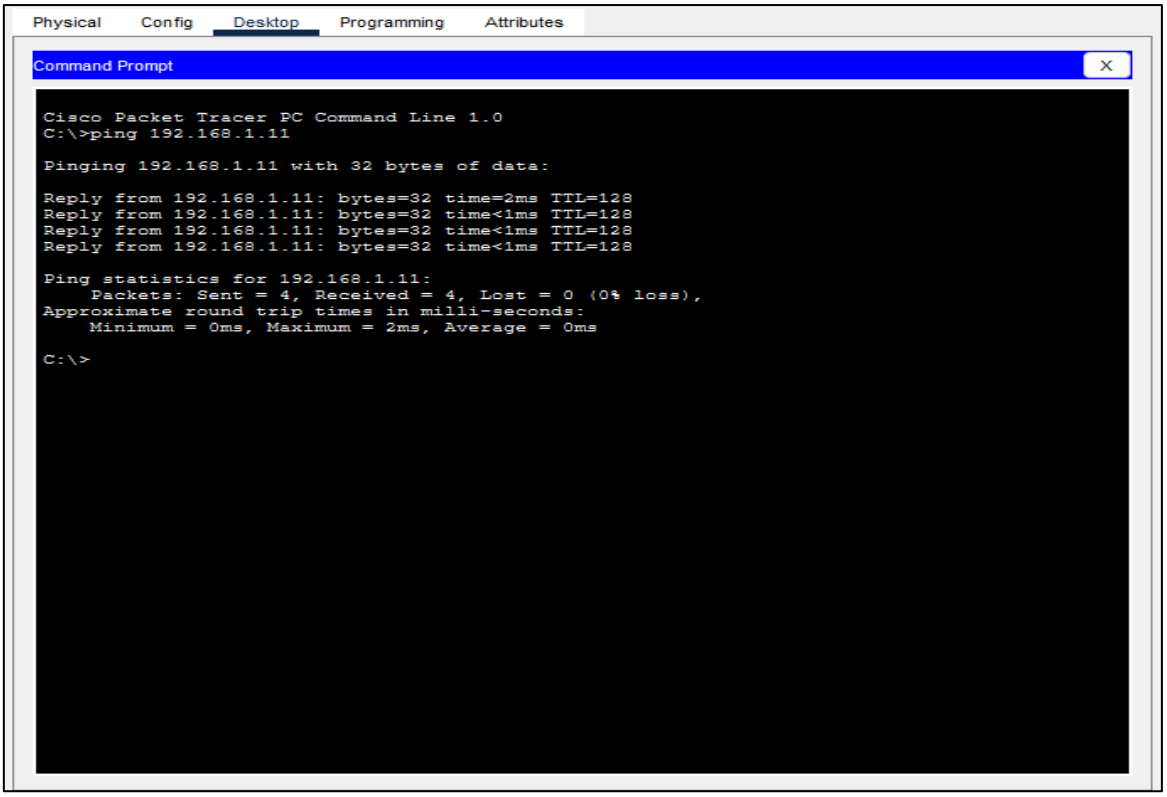
- LAN 1 network: 192.168.1.0/24
- LAN 2 network: 192.168.2.0/24
- Router-to-router network: 10.0.0.0/30
- Static route from Router 1 to LAN 2
- Static route from Router 2 to LAN 1

Output

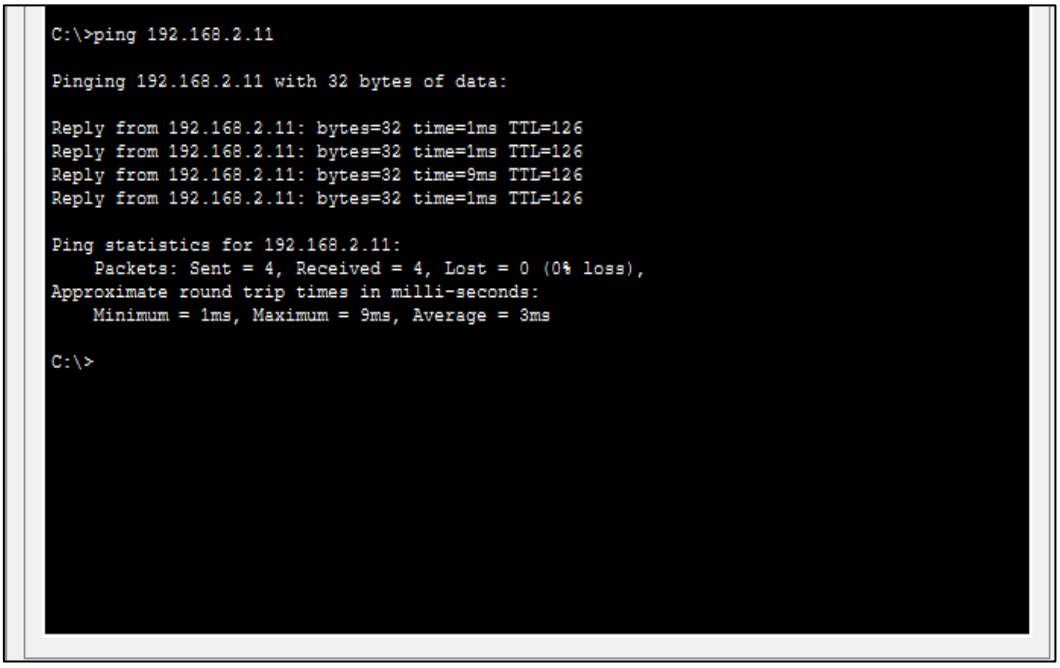
Connectivity



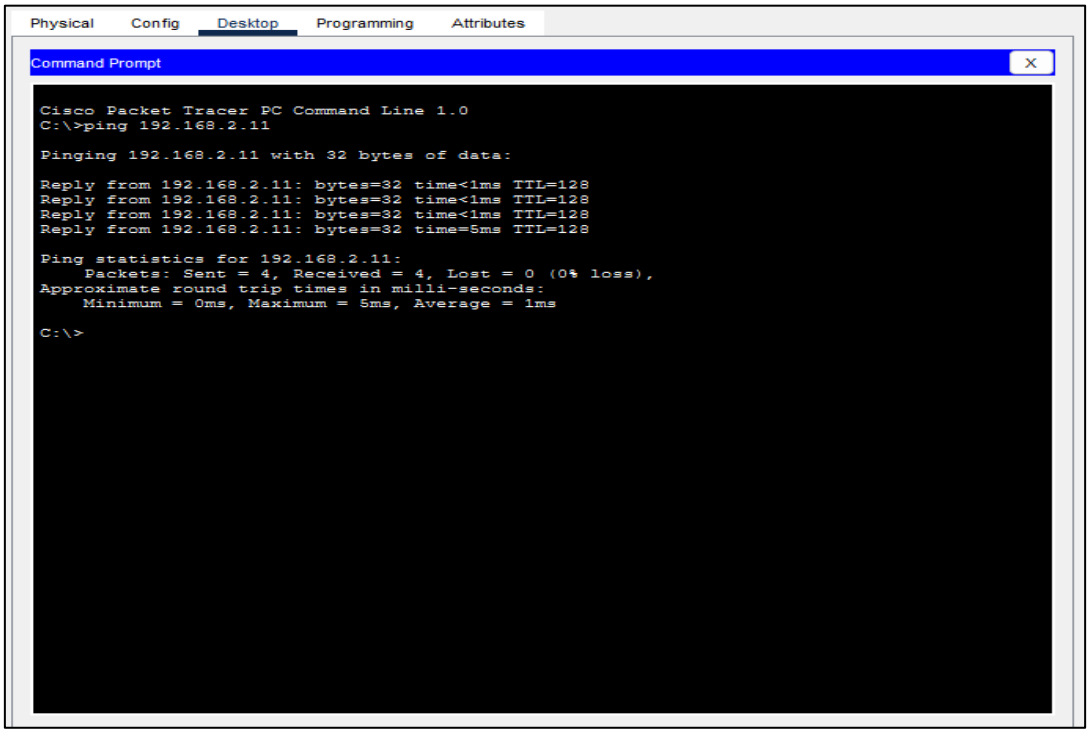
For PC0



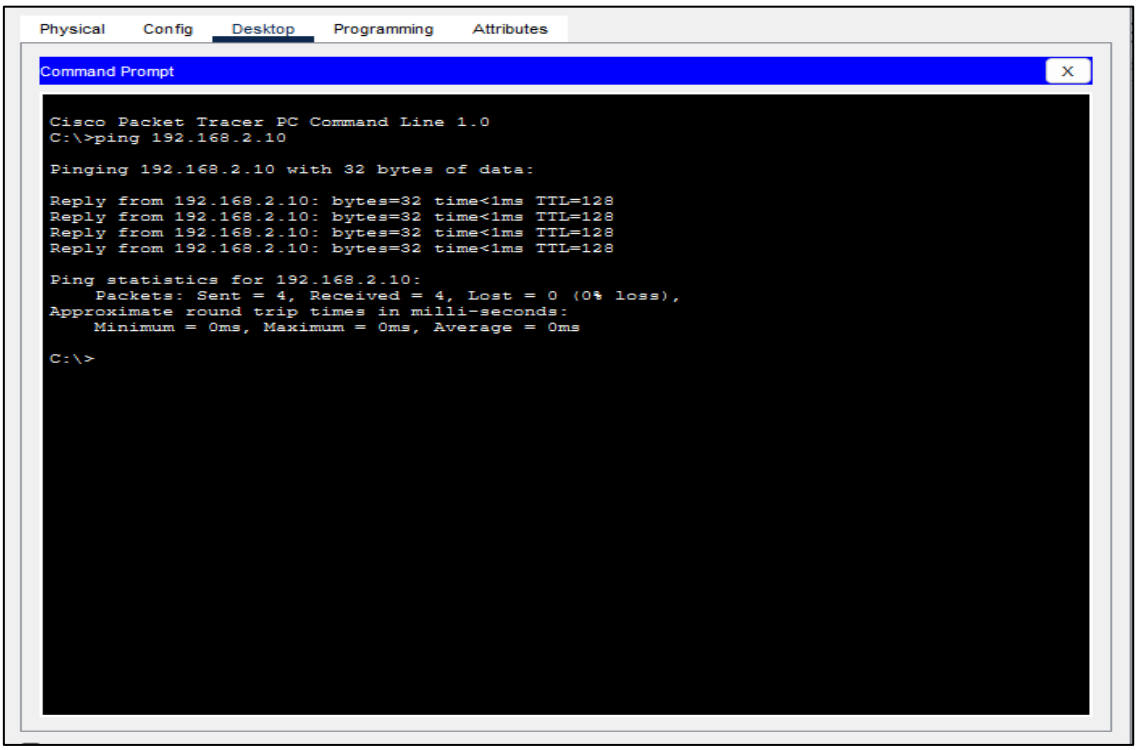
For PC1



For PC2



For PC 3



Result

Two different LANs were successfully connected using two routers, and static routes were configured to establish communication between the LANs. Connectivity was verified successfully using the ping command.

Ex. No: 8 **Name:**

Date: **Reg. No.:**

Multi-Routing Connections with Static Route

Aim

To establish communication between multiple LANs using multiple routers and configure static routing to enable data transmission between all connected networks.

Algorithm

STEP 1: Open Cisco Packet Tracer.

STEP 2: Place three routers, three switches, and PCs.

STEP 3: Connect each group of PCs to a switch.

STEP 4: Connect each switch to a router to form individual LANs.

STEP 5: Connect the routers in a suitable topology.

STEP 6: Assign IP addresses to all PCs and router interfaces.

STEP 7: Configure all router interfaces and enable them using no shutdown.

STEP 8: Configure static routes in each router using the ip route command.

STEP 9: Verify the routing table using show ip route.

STEP 10: Test communication between PCs in different LANs using ping.

STEP 11: Use tracert to observe the path followed by packets.

STEP 12: Record the observations and verify successful communication.

Network Configuration

LAN 1 – Router R1

Device	IP Address	Gateway
PC1	192.168.1.10	192.168.1.1
PC2	192.168.1.11	192.168.1.1

R1 LAN interface:

192.168.1.1 /24

LAN 2 – Router R2

Device	IP Address	Gateway
PC3	192.168.2.10	192.168.2.1
PC4	192.168.2.11	192.168.2.1

R2 LAN interface:

192.168.2.1 /24

LAN 3 – Router R3

Device	IP Address	Gateway
PC5	192.168.3.10	192.168.3.1
PC6	192.168.3.11	192.168.3.1

R3 LAN interface:

192.168.3.1 /24

Router-to-Router Connections

R1 ↔ R2 : 10.0.0.0/30

R2 ↔ R3 : 10.0.0.4/30

Router	Interface	IP Address
R1	R1-R2	10.0.0.1
R2	R1-R2	10.0.0.2
R2	R2-R3	10.0.0.5
R3	R2-R3	10.0.0.6

Subnet Mask: 255.255.255.252

Program / Router Configuration

Router R1

```
Router> enable
Router# configure terminal

Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

```
Router(config)# interface serial 0/0/0
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2 Router(config)# ip
route 192.168.3.0 255.255.255.0 10.0.0.2

Router(config)# end
Router# show ip route
```

Router R2

```
Router> enable
Router# configure terminal

Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface serial 0/0/0
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface serial 0/0/1
Router(config-if)# ip address 10.0.0.5 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1 Router(config)# ip
route 192.168.3.0 255.255.255.0 10.0.0.6

Router(config)# end
```

Router# show ip route

Router R3

Router> enable

Router# configure terminal

Router(config)# interface gigabitEthernet 0/0

Router(config-if)# ip address 192.168.3.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Router(config)# interface serial 0/0/0

Router(config-if)# ip address 10.0.0.6 255.255.255.252

Router(config-if)# no shutdown

Router(config-if)# exit

Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.5 Router(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.5

Router(config)# end

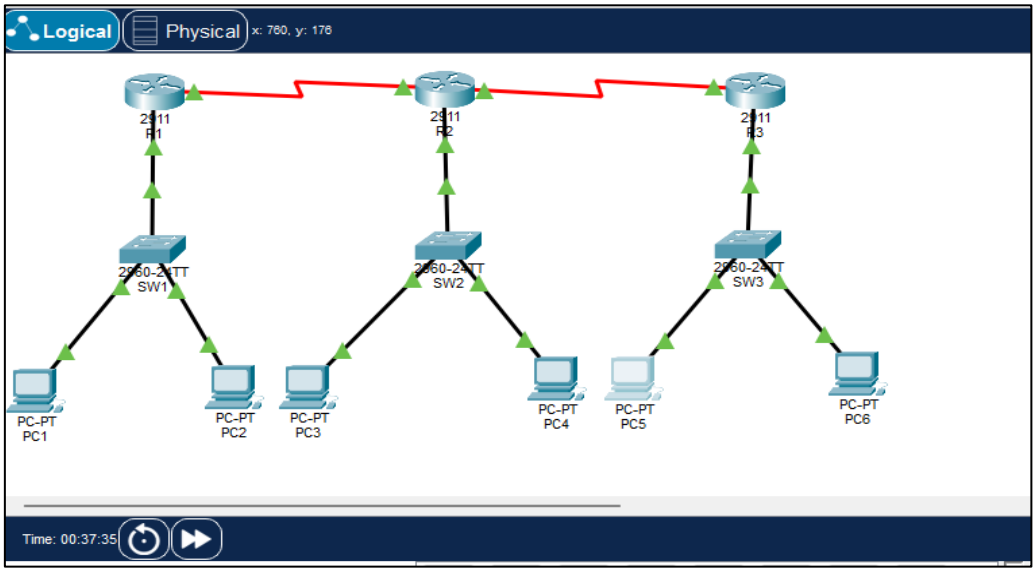
Router# show ip route

Input

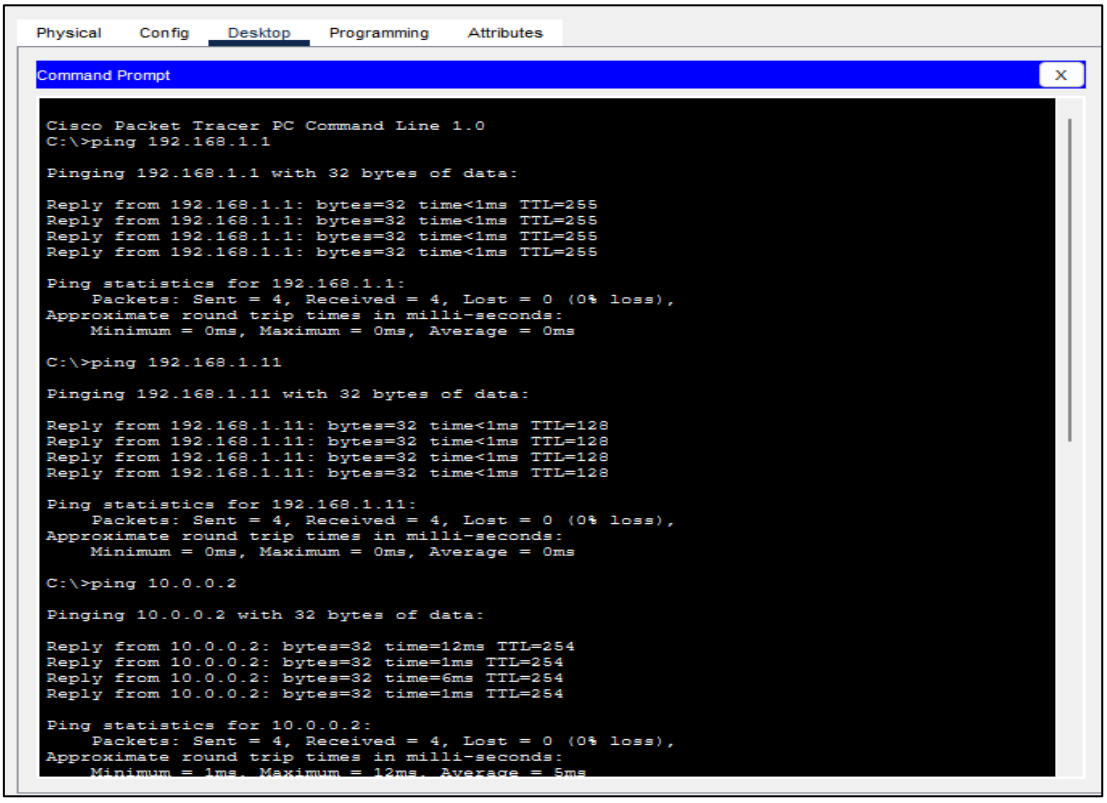
- Number of routers: 3
- Number of LANs: 3
- LAN 1: 192.168.1.0/24
- LAN 2: 192.168.2.0/24
- LAN 3: 192.168.3.0/24
- R1–R2: 10.0.0.0/30
- R2–R3: 10.0.0.4/30
- Routing method: Static Routing

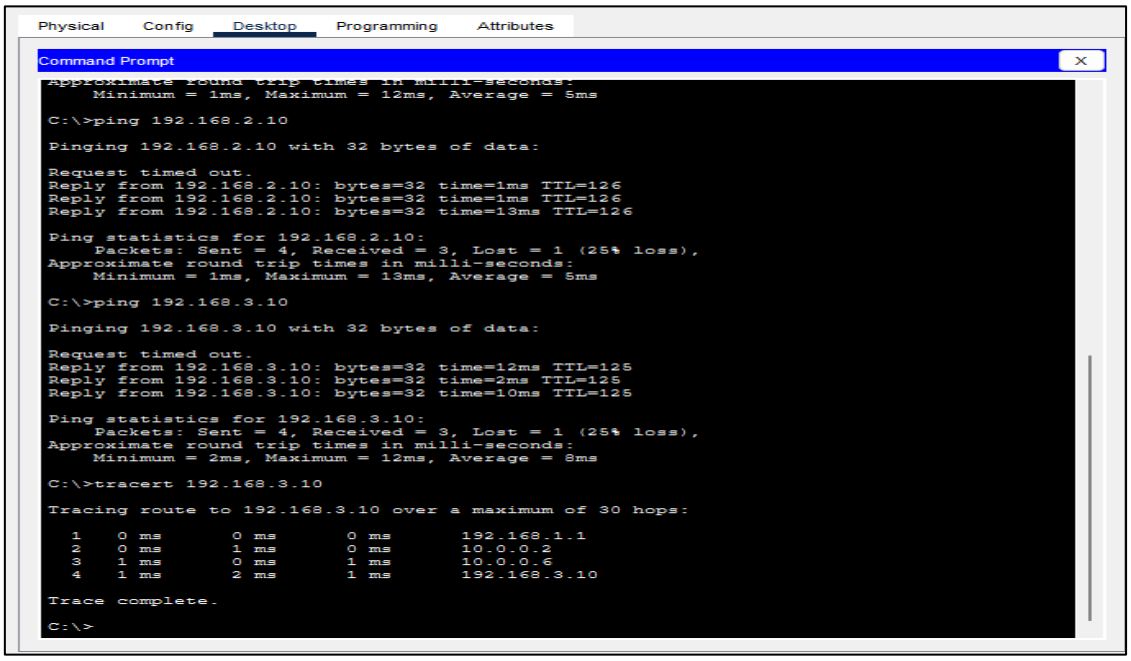
Output

Connectivity

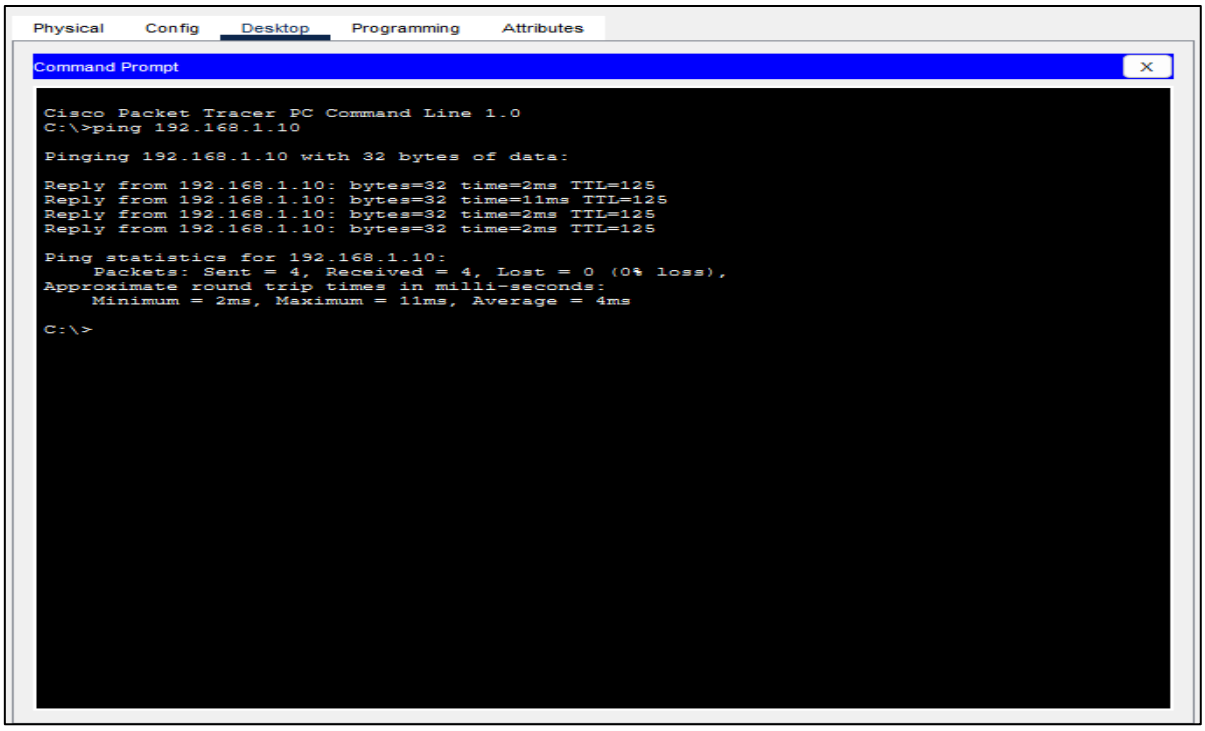


For PC1





For PC2



Result

Multiple LANs were successfully interconnected using three routers, and static routes were configured to provide communication between all networks. The connectivity and routing paths were successfully verified using ping, tracert, and show ip route commands.

Ex. No: 9 **Name:**

Date: **Reg. No.:**

Networking Lab Program: Working With Arp

Aim

To demonstrate the working of ARP (Address Resolution Protocol) and analyze how an IP address is mapped to the corresponding MAC address in a LAN.

Algorithm

STEP 1: Open Cisco Packet Tracer.

STEP 2: Place one switch and two PCs.

STEP 3: Connect both PCs to the switch.

STEP 4: Assign IP addresses to the PCs in the same network.

STEP 5: Open the Command Prompt of PC1.

STEP 6: Check the ARP table using the arp -a command.

STEP 7: Send a ping from PC1 to PC2.

STEP 8: Observe that ARP resolves the IP address of PC2 into its MAC address. Again execute arp -a on PC1.

STEP 9: Observe the newly added IP-to-MAC address entry.

STEP 10: Analyze how ARP enables Ethernet frame delivery.

STEP 11: Record the output and result.

Network Configuration

PC1

IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0

PC2

IP Address : 192.168.1.20
Subnet Mask : 255.255.255.0

Procedure

Step 1: Check ARP Table

On PC1:

PC> arp -a

Initially, the ARP table may contain no entry for PC2.

Step 2: Send Ping

PC> ping 192.168.1.20

The PC sends an ARP Request to find the MAC address associated with 192.168.1.20.

Step 3: Check ARP Table Again

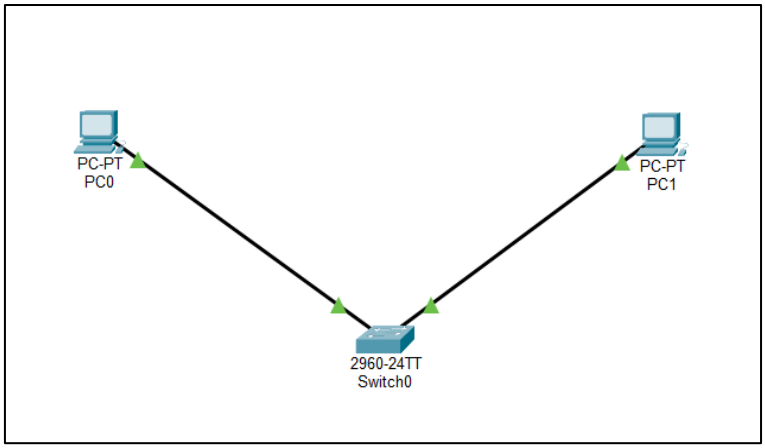
PC> arp -a

Input

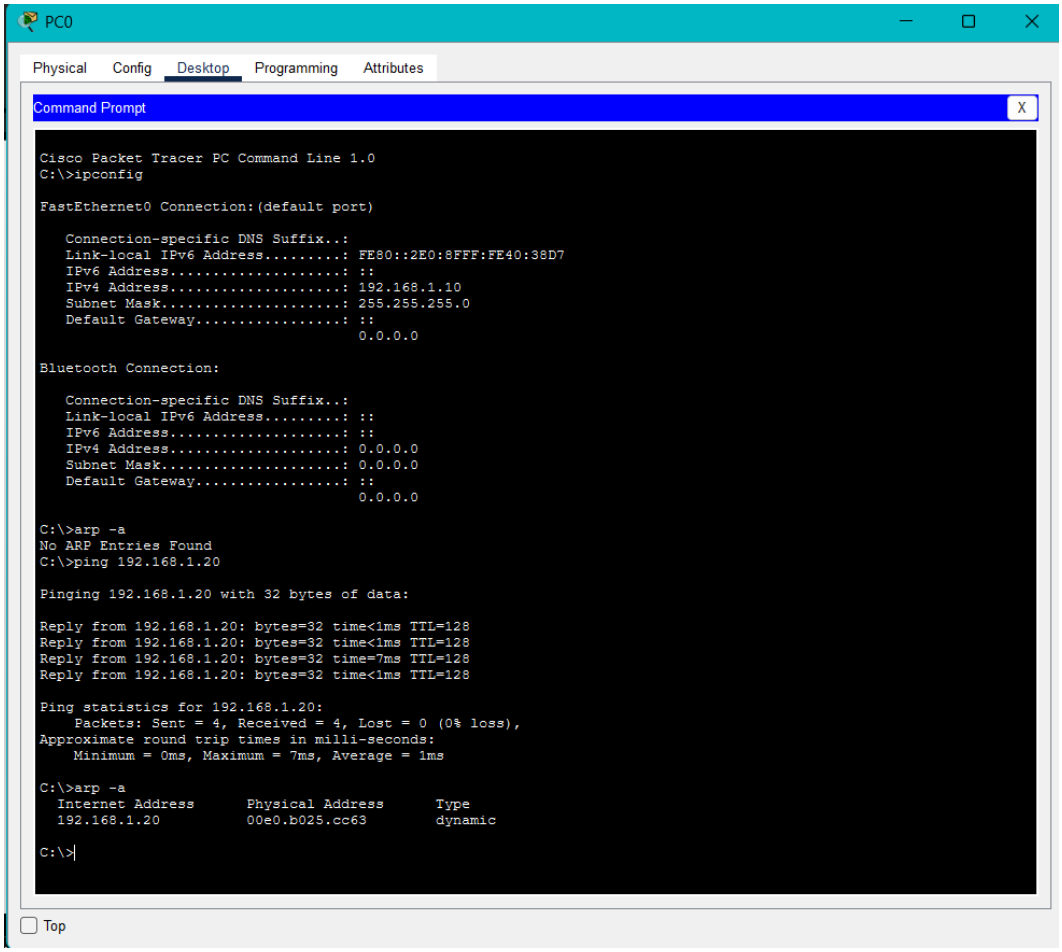
- PC1 IP Address: 192.168.1.10
- PC2 IP Address: 192.168.1.20
- Subnet Mask: 255.255.255.0
- Network: 192.168.1.0/24
- Command: arp -a
- Connectivity command: ping 192.168.1.20

Output

Connectivity



For PC0



Result

The working of ARP was successfully demonstrated using Cisco Packet Tracer. The ARP table was analyzed, and the mapping between the IP address and corresponding MAC address was successfully observed.

Ex. No: 10 **Name:**

Date: **Reg. No.:**

Networking Lab Program: Simulate Working of DNS

Aim

To simulate the working of DNS (Domain Name System) using Cisco Packet Tracer and demonstrate how a domain name is resolved into its corresponding IP address.

Algorithm

STEP 1: Open Cisco Packet Tracer.

STEP 2: Place one PC, one switch, one DNS server, and one web server.

STEP 3: Connect the devices using appropriate cables.

STEP 4: Assign IP addresses to all devices.

STEP 5: Configure the DNS server with a domain name and corresponding IP address.

STEP 6: Configure the PC with the DNS server's IP address.

STEP 7: Open the PC's Command Prompt.

STEP 8: Use the nslookup command to query the domain name.

STEP 9: Observe the IP address returned by the DNS server.

STEP 10: Open the web browser on the PC.

STEP 11: Enter the configured domain name.

STEP 12: Verify that the web page is accessed using DNS name resolution.

STEP 13: Record the output and result.

Network Configuration

PC

IP Address : 192.168.1.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

DNS Server : 192.168.1.2

DNS Server

IP Address : 192.168.1.2

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

Web Server

IP Address : 192.168.1.3

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

Program / Commands

From the PC Command Prompt:

PC> nslookup www.example.com

Test connectivity:

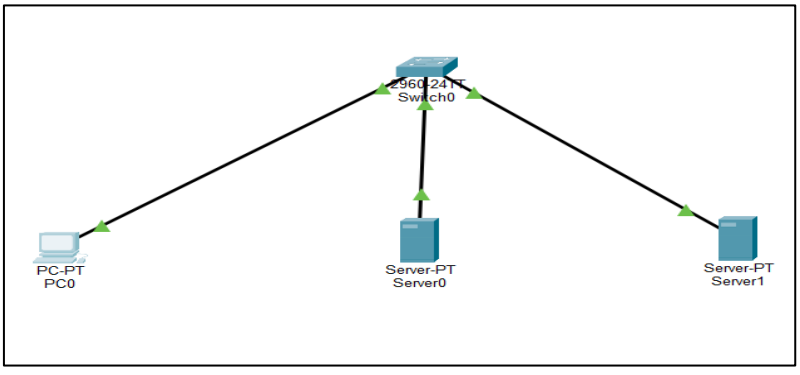
PC> ping www.example.com

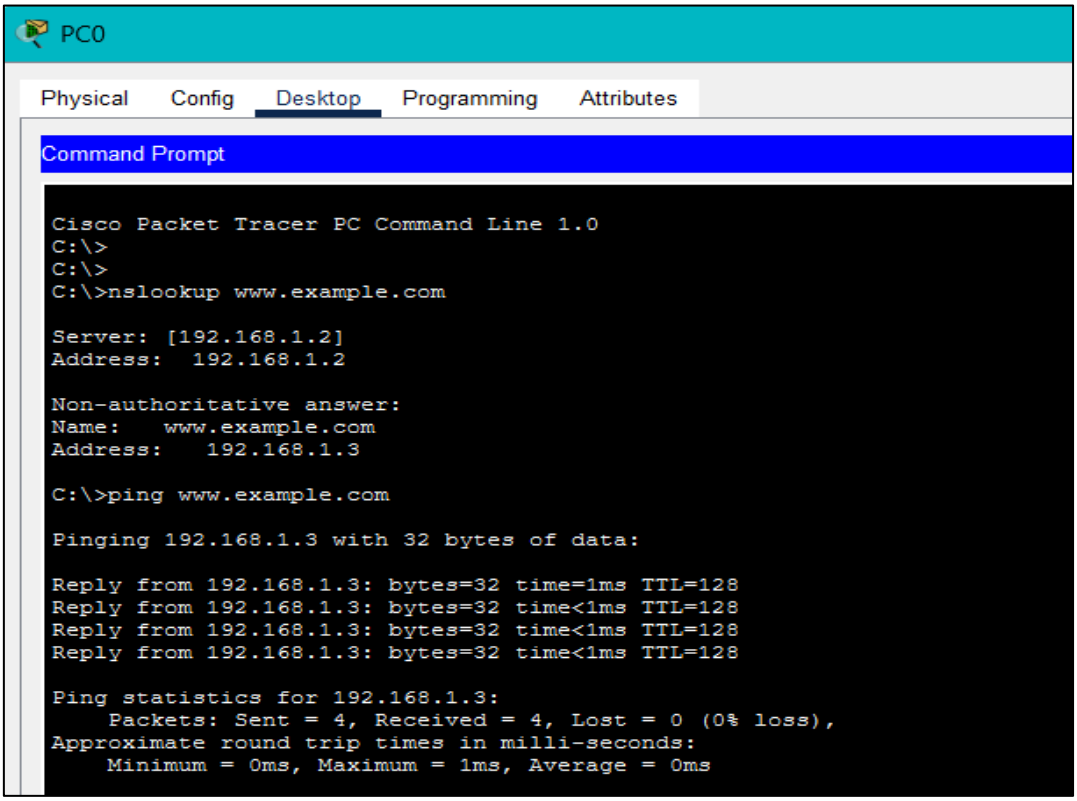
Input

- PC IP: 192.168.1.10
- DNS Server IP: 192.168.1.2
- Web Server IP: 192.168.1.3
- Domain Name: www.example.com
- DNS Record: www.example.com → 192.168.1.3

Output

Connectivity





Result

The working of DNS was successfully simulated using Cisco Packet Tracer. The domain name www.example.com was successfully resolved to its corresponding IP address using the configured DNS server, and communication with the web server was verified.

Ex. No: 11 **Name:**

Date: **Reg. No.:**

Networking Lab Program: Simulate Working of FTP

Aim

To simulate the working of FTP (File Transfer Protocol) using Cisco Packet Tracer and demonstrate the transfer of files between a client and an FTP server.

Algorithm

STEP 1: Open Cisco Packet Tracer.

STEP 2: Place one PC, one switch, and one server.

STEP 3: Connect the PC and server to the switch using appropriate cables.

STEP 4: Assign IP addresses to the PC and FTP server.

STEP 5: Configure the server as an FTP server.

STEP 6: Enable the FTP service on the server.

STEP 7: Create an FTP username and password.

STEP 8: Open the Command Prompt on the PC.

STEP 9: Establish a connection to the FTP server using the ftp command.

STEP 10: Enter the configured username and password.

STEP 11: Use FTP commands such as dir, get, and put to access and transfer files.

STEP 12: Observe the successful communication and file transfer.

STEP 13: Record the input, output, and result.

Network Configuration

PC – FTP Client

IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

FTP Server

IP Address : 192.168.1.20
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

FTP Server Configuration

In Server → Services → FTP:

- Turn FTP Service: ON
- Create a user account:

Username : student
Password : student123

Procedure

From the PC Command Prompt:

Step 1: Test Connectivity

PC> ping 192.168.1.20

Step 2: Connect to FTP Server

PC> ftp 192.168.1.20
Enter the credentials:
Username: student
Password: student123

Step 3: Display Files

ftp> dir

Step 4: Download a File

ftp> get c2960-lanbase-mz.122-25.FX.bin
Transfer complete - 4441921 bytes
4441921 bytes copied in 24.02 secs

Step 5: Close the FTP Connection

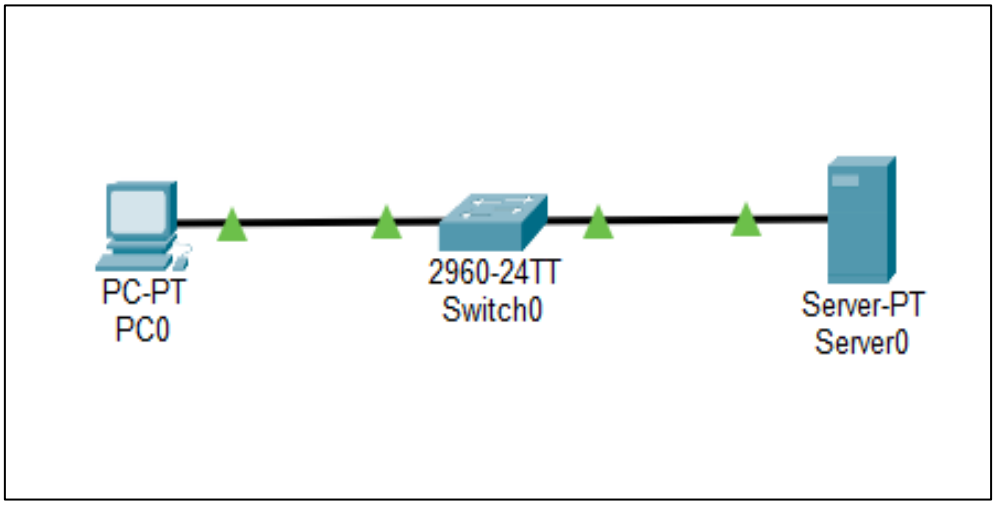
ftp> quit

Input

- FTP Client IP: 192.168.1.10
- FTP Server IP: 192.168.1.20
- Username: student
- Password: student123
- File: c2960-lanbase-mz.122-25.FX.bin
- FTP commands: dir, get, quit

Output

Connectivity



Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ftp 192.168.1.20
Trying to connect...192.168.1.20
Connected to 192.168.1.20
220- Welcome to PT Ftp server
Username:student
331- Username ok, need password
Password:**********
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.1.20:
0    : asa842-k8.bin                                  5571584
1    : asa923-k8.bin                                  30468096
2    : c1841-advipservicesk9-mz.124-15.T1.bin         33591768
3    : c1841-ipbase-mz.123-14.T7.bin                  13832032
4    : c1841-ipbasek9-mz.124-12.bin                   16599160
5    : c1900-universalk9-mz.SPA.155-3.M4a.bin         33591768
6    : c2600-advipservicesk9-mz.124-15.T1.bin         33591768
7    : c2600-i-mz.122-28.bin                          5571584
8    : c2600-ipbasek9-mz.124-8.bin                    13169700
9    : c2800nm-advipservicesk9-mz.124-15.T1.bin       50938004
10   : c2800nm-advipservicesk9-mz.151-4.M4.bin        33591768
11   : c2800nm-ipbase-mz.123-14.T7.bin                5571584
12   : c2800nm-ipbasek9-mz.124-8.bin                  15522644
13   : c2900-universalk9-mz.SPA.155-3.M4a.bin         33591768
14   : c2950-i6q412-mz.121-22.EA4.bin                 3058048
15   : c2950-i6q412-mz.121-22.EA8.bin                 3117390
16   : c2960-lanbase-mz.122-25.FX.bin                 4414921
17   : c2960-lanbase-mz.122-25.SEE1.bin               4670455
```

```
18  : c2960-lanbasek9-mz.150-2.SE4.bin                  4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin           8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin            10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin              33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin             83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin            505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG               159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG               184530138
26  : ie9k_iosxe.17.09.04.SPA.bin                       596133776
27  : ir800-universalk9-bundle.SPA.156-3.M.bin          160968869
28  : ir800-universalk9-mz.SPA.155-3.M                  61750062
29  : ir800-universalk9-mz.SPA.156-3.M                  63753767
30  : ir800_yocto-1.7.2.tar                             2877440
31  : ir800_yocto-1.7.2_python-2.7.3.tar                6912000
32  : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg     685645824
33  : pt1000-i-mz.122-28.bin                            5571584
34  : pt3000-i6q4l2-mz.121-22.EA4.bin                   3117390
ftp> get sample.txt

Reading file sample.txt from 192.168.1.20:
File transfer in progress...

%Error ftp://192.168.1.20/sample.txt (No such file or directory Or Permission denied)
550-Requested action not taken. File unavailable (e.g., file not found).

ftp>get c2960-lanbase-mz.122-25.FX.bin

Reading file c2960-lanbase-mz.122-25.FX.bin from 192.168.1.20:
File transfer in progress...

[Transfer complete - 4414921 bytes]

4414921 bytes copied in 24.024 secs (4993 bytes/sec)
ftp>quit

221- Service closing control connection.
```

Result

The working of FTP was successfully simulated using Cisco Packet Tracer. The FTP client established a connection with the FTP server, authenticated successfully, and performed file upload and download operations.

Ex. No: 12 **Name:**

Date: **Reg. No.:**

Networking Lab Program: Simulate Congestion Control

Aim

To simulate network congestion control and observe how the transmission rate is controlled when network traffic increases.

Algorithm

STEP 1: Open Cisco Packet Tracer.

STEP 2: Create a network consisting of two PCs, two routers, and a server.

STEP 3: Connect the devices using appropriate links.

STEP 4: Assign IP addresses to all devices.

STEP 5: Configure the router interfaces and enable them.

STEP 6: Generate normal traffic between the source and destination.

STEP 7: Increase the amount of traffic transmitted through the network.

STEP 8: Observe packet transmission and possible packet loss or delay.

STEP 9: Reduce the transmission rate when congestion occurs.

STEP 10: Observe the improvement in packet delivery.

STEP 11: Compare network performance before and after congestion control.

STEP 12: Record the observations and result.

Network Configuration

PC1 – Source

IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Router R1

LAN Interface : 192.168.1.1
WAN Interface : 10.0.0.1

Router R2

WAN Interface : 10.0.0.2
LAN Interface : 192.168.2.1

PC2 – Destination

IP Address : 192.168.2.10
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.2.1

Procedure

Step 1: Verify Connectivity

From PC1:
PC> ping 192.168.2.10

Step 2: Generate Continuous Traffic

PC> ping 192.168.2.10 -t
This generates continuous packets between the source and destination.

Step 3: Stop Traffic

Ctrl + C

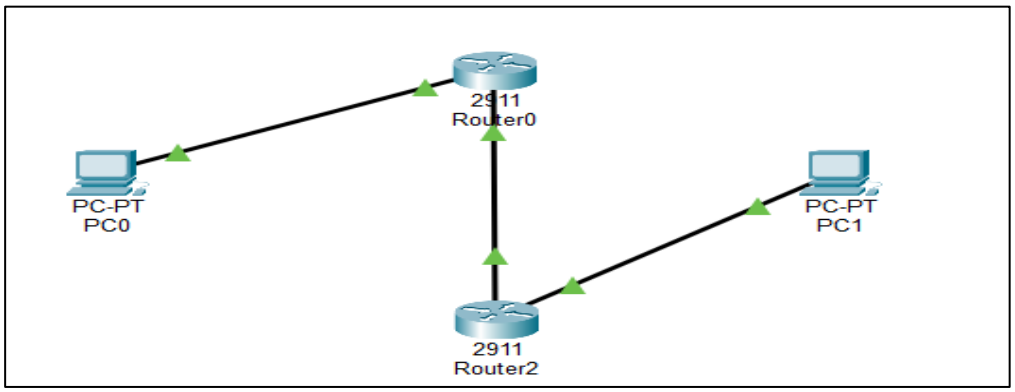
Step 4: Check Router Interfaces

Input

- Source PC: 192.168.1.10
- Destination PC: 192.168.2.10
- Router R1: 192.168.1.1 / 10.0.0.1
- Router R2: 10.0.0.2 / 192.168.2.1
- Traffic: Continuous ICMP packets
- Congestion condition: Increased packet transmission

Output

Connectivity



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=14ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping -t 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 8, Received = 8, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Result

The congestion control mechanism was successfully simulated by generating varying levels of network traffic. The effect of increased traffic on packet delay and packet loss was observed, and the importance of controlling the transmission rate to reduce network congestion was demonstrated.

Ex. No: 13 **Name:**

Date: **Reg. No.:**

Networking Lab Program: Simulate the Working of DHCP

Aim

To simulate the working of DHCP (Dynamic Host Configuration Protocol) using Cisco Packet Tracer and demonstrate automatic assignment of IP addresses to client devices.

Algorithm

STEP 1: Open Cisco Packet Tracer.

STEP 2: Place one DHCP Server, one switch, and multiple PCs.

STEP 3: Connect the PCs and DHCP Server to the switch.

STEP 4: Assign a static IP address to the DHCP Server.

STEP 5: Enable the DHCP service on the server.

STEP 6: Create a DHCP pool by specifying the network address, subnet mask, default gateway, and DNS server.

STEP 7: Configure the PCs to obtain their IP addresses automatically using DHCP.

STEP 8: Verify the IP address assigned to each PC.

STEP 9: Use the ipconfig command to display the assigned configuration.

STEP 10: Use the ping command to verify communication between the PCs and server.

STEP 11: Observe the DHCP address allocation.

STEP 12: Record the input, output, observation and result.

Network Configuration

DHCP Server

IP Address : 192.168.1.2
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

DHCP Pool

Configure the following under:
Server → Services → DHCP → ON
Pool Name : LAN_POOL
Default Gateway : 192.168.1.1
DNS Server : 192.168.1.2
Start IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0
Maximum Users : 50

Client Configuration

On each PC:
Desktop → IP Configuration → DHCP
The PC will automatically request an IP address from the DHCP server.

Program / Commands

Check the IP configuration:

```
PC> ipconfig
```

Test connectivity with the DHCP server:

```
PC> ping 192.168.1.2
```

Test communication with another PC:

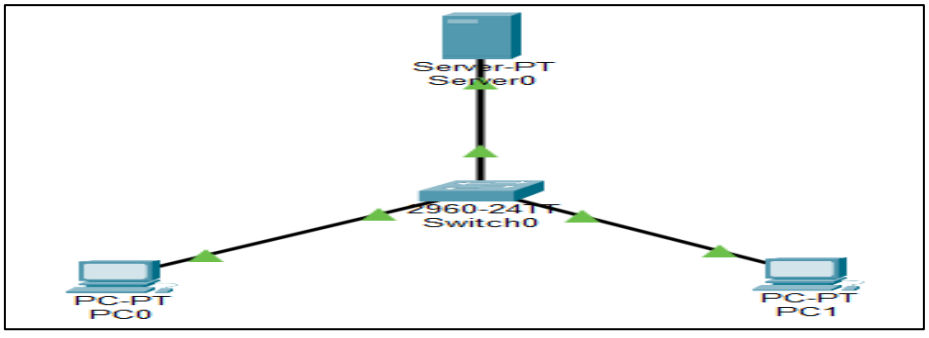
```
PC> ping 192.168.1.11
```

Input

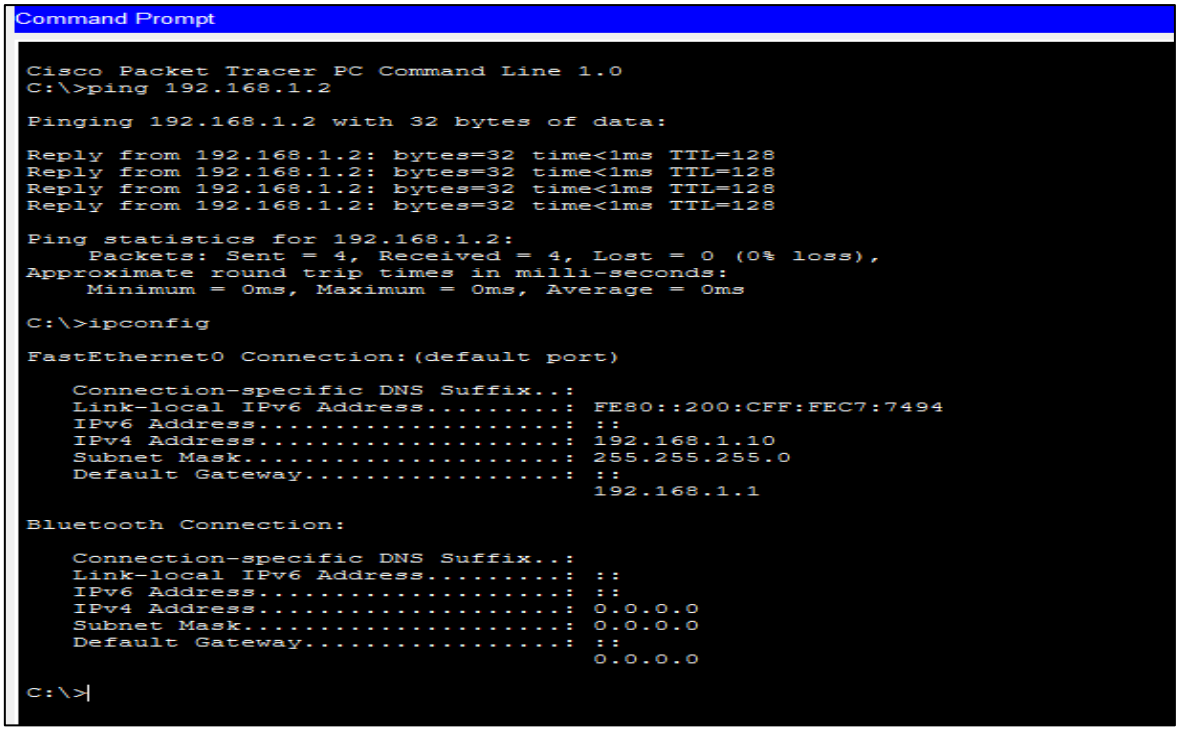
- DHCP Server IP: 192.168.1.2
- Network: 192.168.1.0/24
- Default Gateway: 192.168.1.1
- Starting IP: 192.168.1.10
- Maximum Users: 50
- Client configuration: DHCP

Output

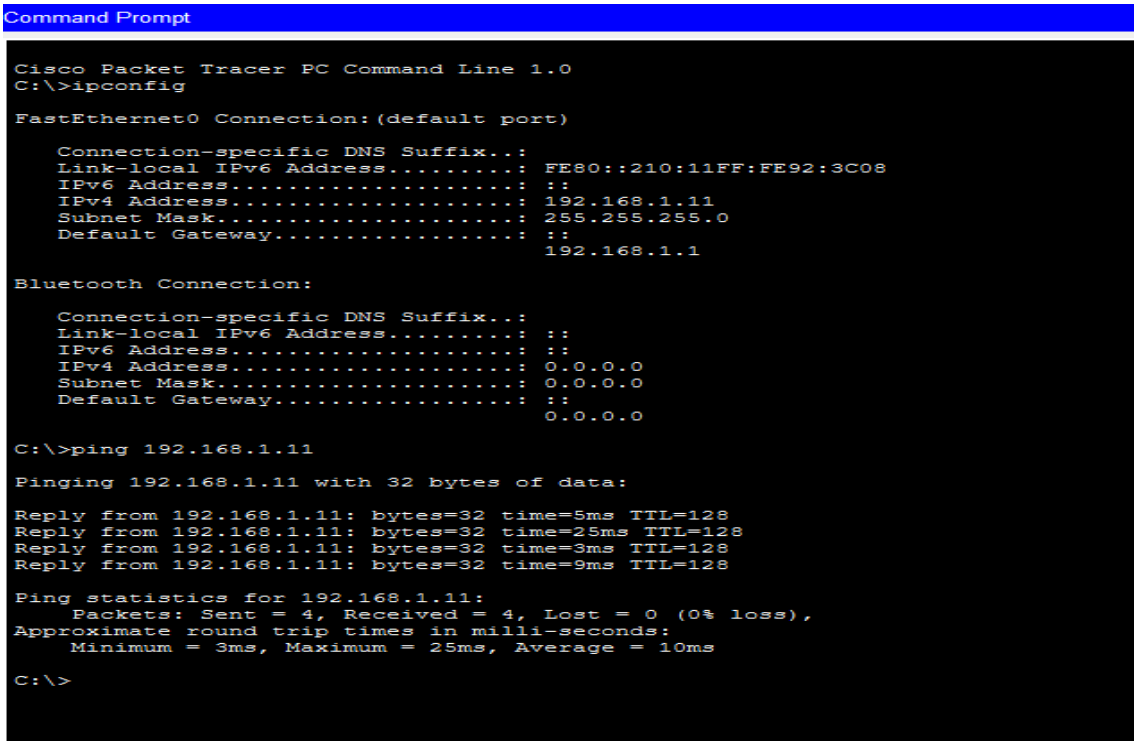
Connectivity



For PC0



For PC1



Result

The working of DHCP was successfully simulated using Cisco Packet Tracer. The DHCP server automatically assigned IP addresses, subnet masks, default gateway, and DNS information to the client PCs, and network connectivity was successfully verified.

Ex. No: 14 **Name:**

Date: **Reg. No.:**

SMTP Simulation

Aim

To simulate the working of SMTP (Simple Mail Transfer Protocol) using Cisco Packet Tracer and demonstrate sending an email from one client to another through a mail server.

Algorithm

STEP 1: Open Cisco Packet Tracer.

STEP 2: Place two PCs, one switch, and one server.

STEP 3: Connect the PCs and server to the switch.

STEP 4: Assign IP addresses to all devices.

STEP 5: Configure the server as a Mail Server.

STEP 6: Enable the SMTP service on the server.

STEP 7: Create email accounts for the sender and receiver.

STEP 8: Configure the email application on both PCs.

STEP 9: Compose an email from the sender PC.

STEP 10: Enter the receiver's email address and send the message.

STEP 11: The SMTP server processes the outgoing email.

STEP 12: Open the receiver's email application and check for the received message.

STEP 13: Verify successful email communication.

STEP 14: Record the input, output, and result.

Network Configuration

Mail Server

IP Address : 192.168.1.2
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Enable

Services → EMAIL → SMTP → ON

Set the domain name:
Domain Name: example.com
Create user accounts:
Username : student1
Password : 12345
Username : student2
Password : 12345

PC1 – Sender Configuration:
IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Email configuration:
Email Address : student1@example.com
Incoming Server : 192.168.1.2
Outgoing Server : 192.168.1.2
Username : student1
Password : 12345

PC2 – Receiver Configuration:
IP Address : 192.168.1.11
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Email configuration:
Email Address : student2@example.com
Incoming Server : 192.168.1.2
Outgoing Server : 192.168.1.2
Username : student2
Password : 12345

Program / Commands

Test connectivity from PC1:
PC> ping 192.168.1.2
The actual SMTP operation is performed through the Email application in Cisco Packet Tracer.

On PC1

Desktop → Email → Compose
Enter:
To: student2@example.com
Subject : Test Mail

Message : Hello, this is an SMTP simulation.
Click Send.

On PC2

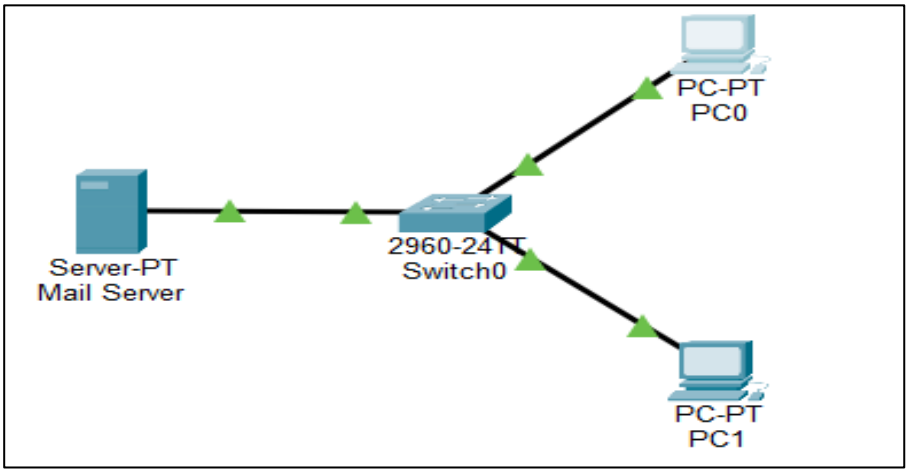
Desktop → Email → Receive

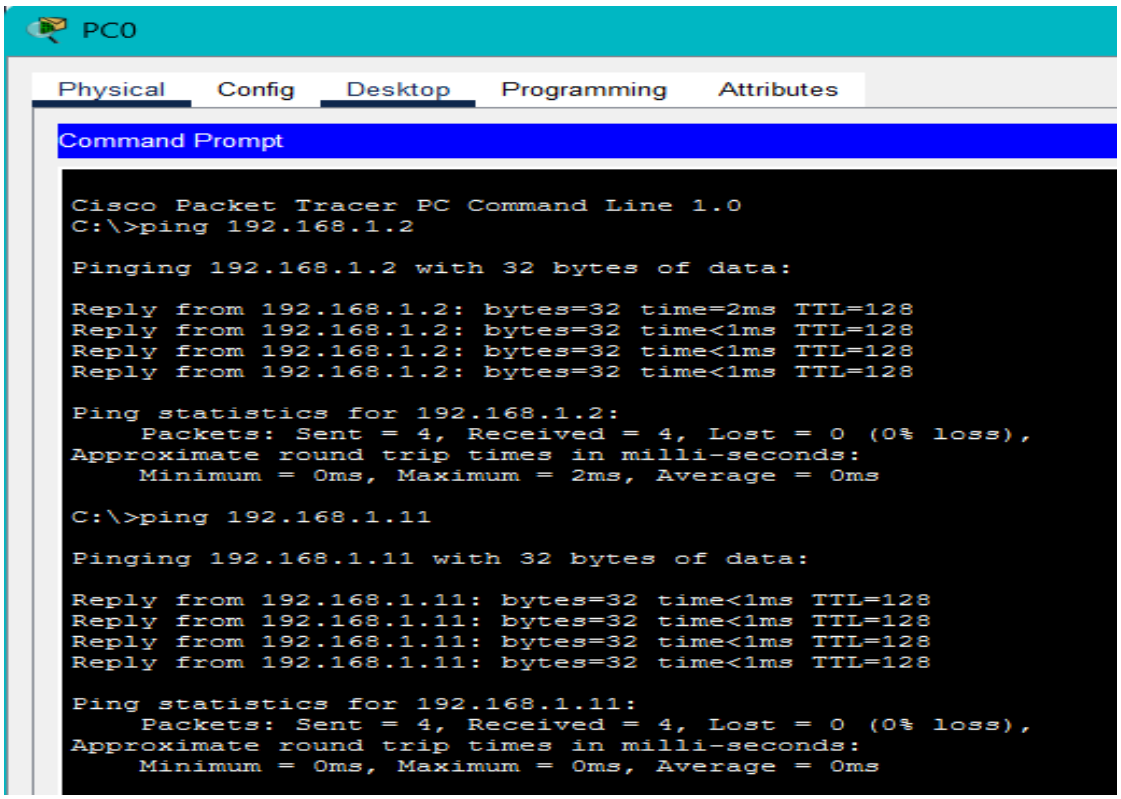
Input

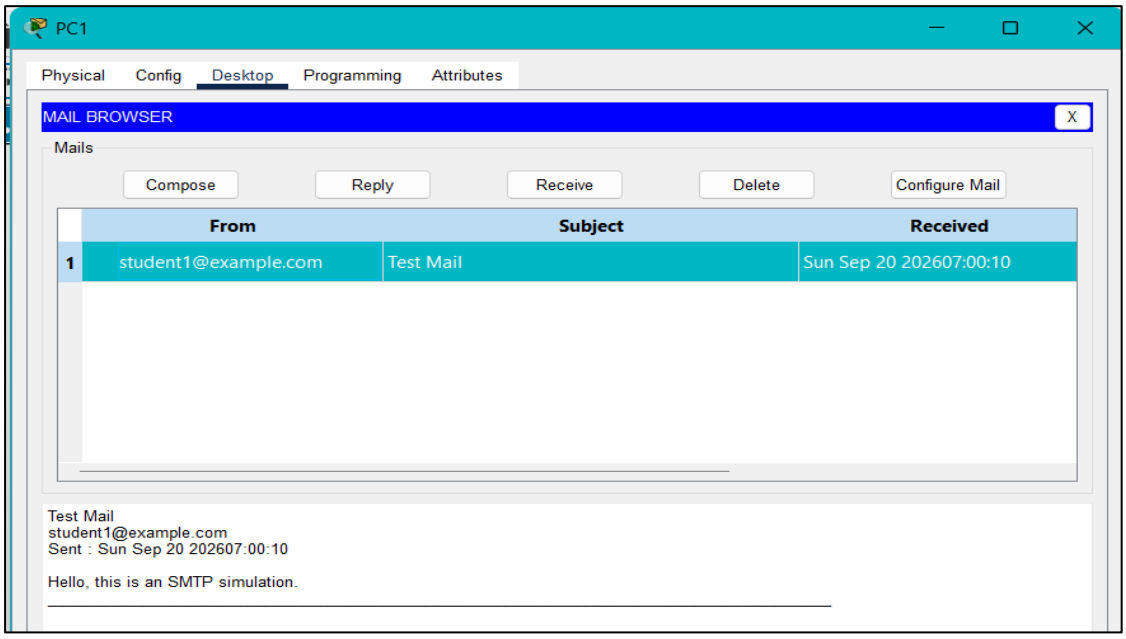
- Mail Server IP: 192.168.1.2
- Sender: student1@example.com
- Receiver: student2@example.com
- SMTP Server: 192.168.1.2
- Subject: Test Mail
- Message: Hello, this is an SMTP simulation.

Output

Connectivity







Result

The working of SMTP was successfully simulated using Cisco Packet Tracer. An email was successfully sent from the sender to the receiver through the configured mail server, demonstrating the basic working of SMTP.

Ex. No: 15 **Name:**

Date: **Reg. No.:**

SDG Mini Project

Aim

To design and simulate a Smart Waste Management System using Cisco Packet Tracer and IoT components to monitor waste levels and identify when a waste bin reaches the predefined full-level threshold.

TITLE: Smart Waste Management System Using IoT

SDG Mapping

- **Primary SDG:** SDG 11 – Sustainable Cities and Communities
- **Supporting SDG:** SDG 12 – Responsible Consumption and Production
- **Supporting SDG:** SDG 13 – Climate Action

Problem Statement

Traditional waste collection systems may result in overflowing bins when waste levels are not monitored properly. A smart waste management system can monitor the waste level and identify when the bin reaches a predefined threshold, helping to indicate when waste collection is required.

Objectives

1. To monitor the waste level of a simulated garbage bin.
2. To represent the waste level using a potentiometer.
3. To process the waste-level information using the MCU-PT.
4. To compare the waste level with a predefined threshold of 80%.
5. To identify the bin as NORMAL or FULL.
6. To indicate when waste collection is required.
7. To demonstrate an IoT-based approach to smart waste management.

System Requirements

Hardware / Cisco Components

- MCU-PT
- Potentiometer (IoT0)
- LED (IoT1)

Software

- Cisco Packet Tracer
- MCU-PT Programming Environment
- Python 2 program in the MCU

Algorithm

STEP 1: Start the system.

STEP 2: Set the bin capacity to 100%.

STEP 3: Set the full-level limit to 80%.

STEP 4: Connect the potentiometer to the MCU-PT to represent the waste-level input.

STEP 5: Connect the LED to the MCU-PT to represent the bin-status indicator.

STEP 6: Define the test waste levels as 50%, 70%, 80%, and 85%.

STEP 7: Take each waste level one by one.

STEP 8: Display the current waste level.

STEP 9: Compare the waste level with the 80% threshold.

STEP 10: If the waste level is 80% or above, display BIN FULL.

STEP 11: Display Waste collection required.

STEP 12: If the waste level is below 80%, display BIN NORMAL.

STEP 13: Display No immediate collection required.

STEP 14: Repeat the process for all the predefined waste levels.

STEP 15: Stop the system.

Program

```
# Smart Waste Management - Basic Simulation
BIN_CAPACITY = 100
FULL_LEVEL = 80

waste_levels = [50, 70, 80, 85]

print("SMART WASTE MANAGEMENT SYSTEM")
print("------------------------------")
```

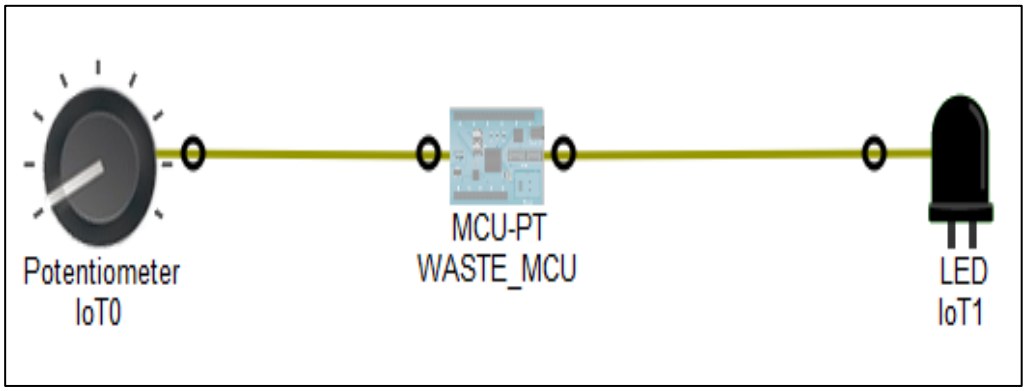
```
for waste_level in waste_levels:
   print("Waste Level : %d%%" % waste_level)

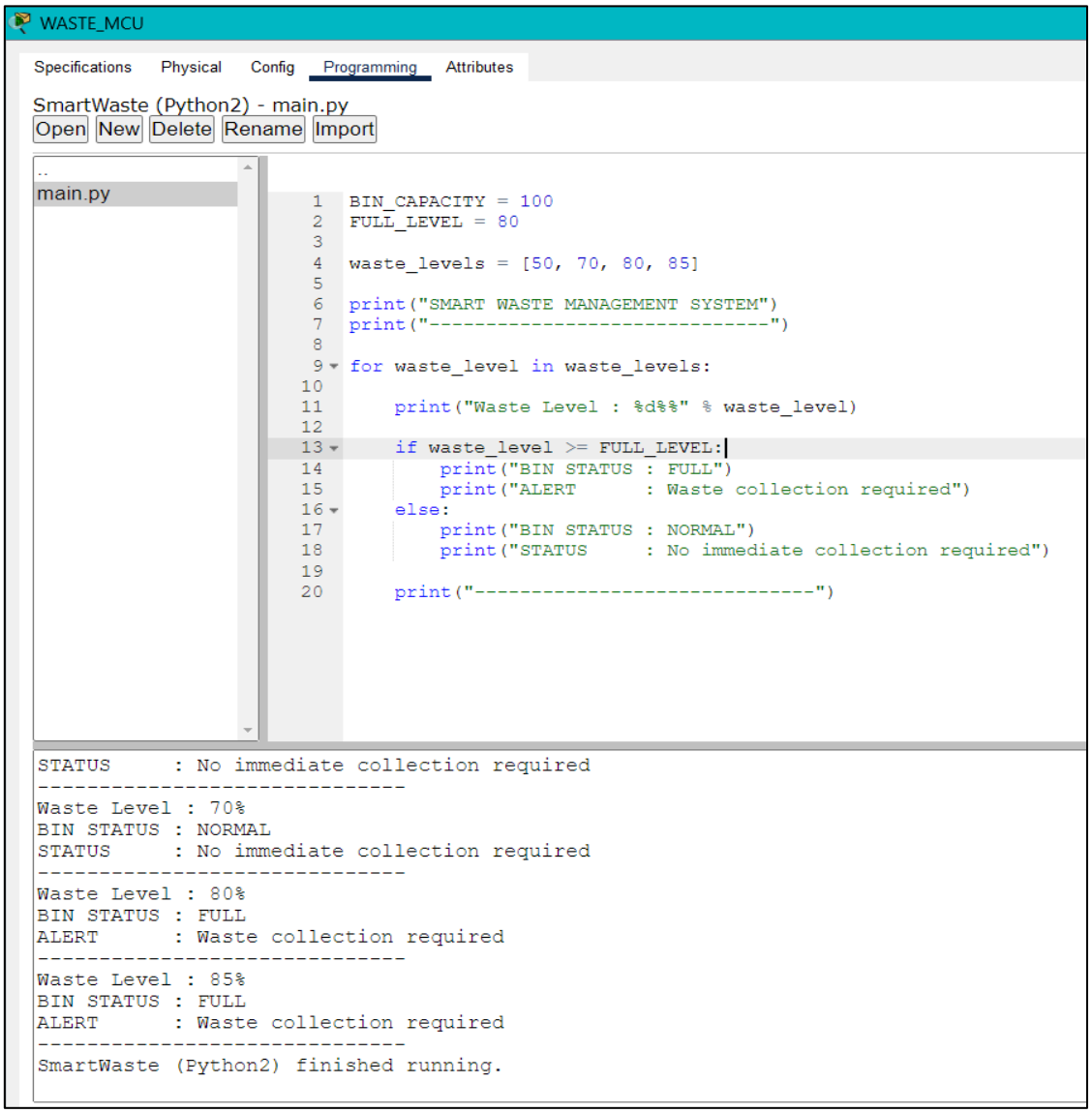
if waste_level >= FULL_LEVEL:
      print("BIN STATUS : FULL")
      print("ALERT     : Waste collection required")
else:
      print("BIN STATUS : NORMAL")
      print("STATUS    : No immediate collection required")

print("------------------------------")
```

Output

Connectivity





Result

The Smart Waste Management System was successfully simulated in Cisco Packet Tracer using an MCU-PT, potentiometer, and LED. The system tested waste levels of 50%, 70%, 80%, and 85%. Below 80% was shown as **BIN NORMAL**, while 80% or above was shown as **BIN FULL** with a collection alert.